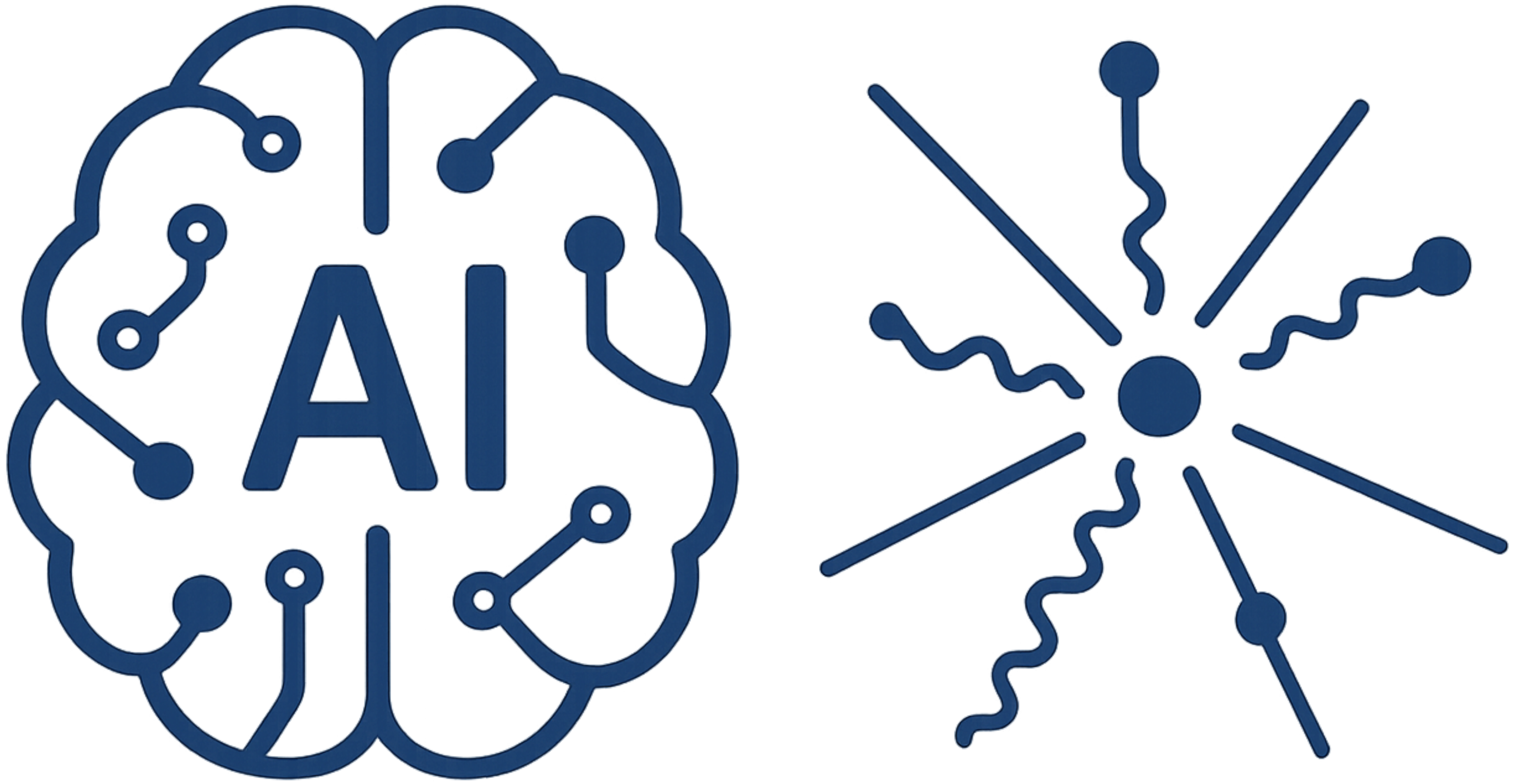
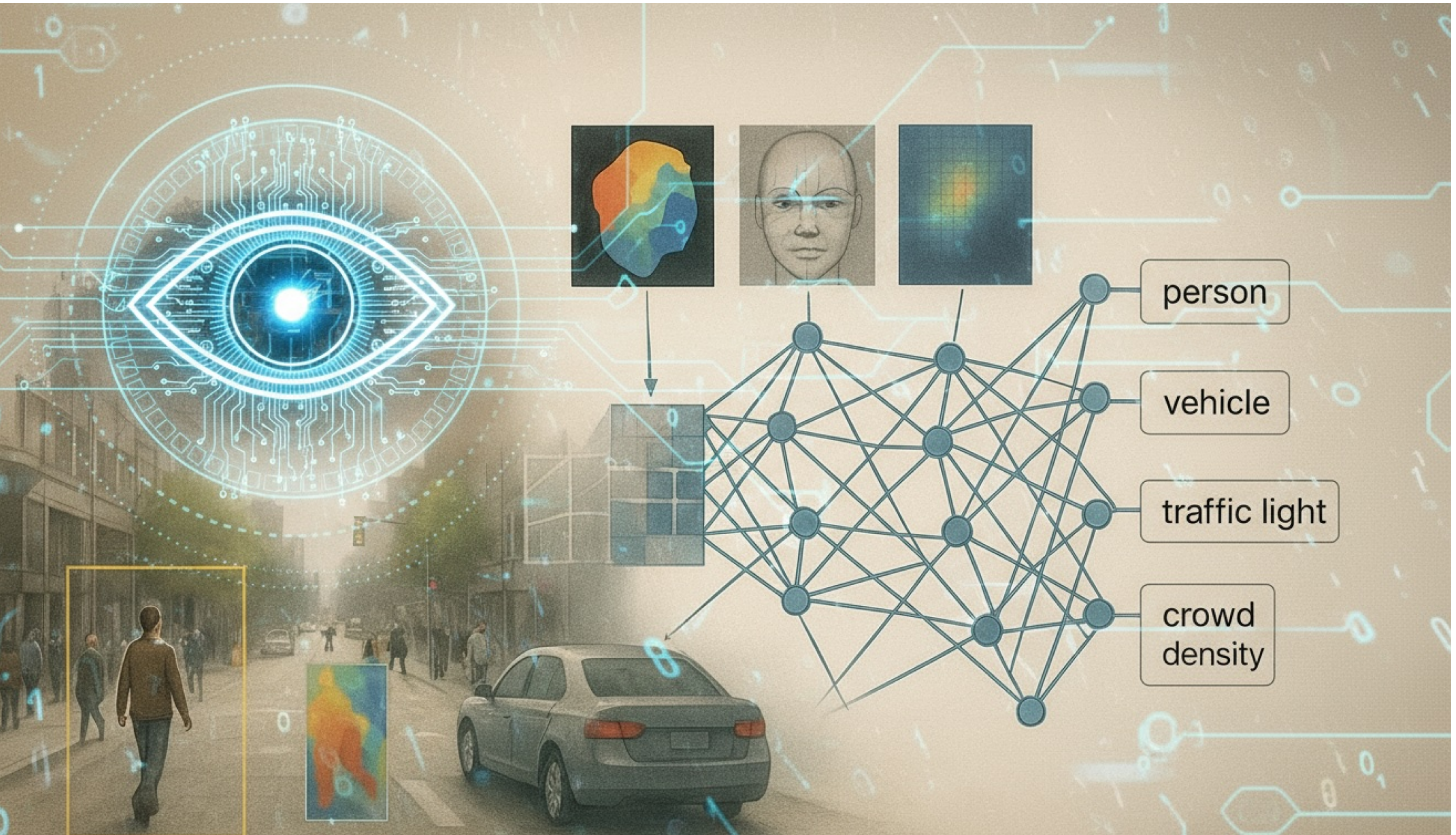
Introduction to

AI-Driven HEP

S. A. Fard - School of Physics (IPM)



AI-Driven HEP 14



Session 14

Computer Vision III

Reference

[1] Bishop, Christopher M., and Hugh Bishop. Deep learning: Foundations and concepts. Springer Nature, 2023.

[2] James, Gareth, et al. "Statistical learning." An introduction to statistical learning: With applications in Python. (2023)

[3] Foundations of Computer Vision, https://visionbook.mit.edu/

Optimum Problem

How many parameters we need to explain a population well ?

Optimum Problem

Consider a box

$1\ m^3$ volume

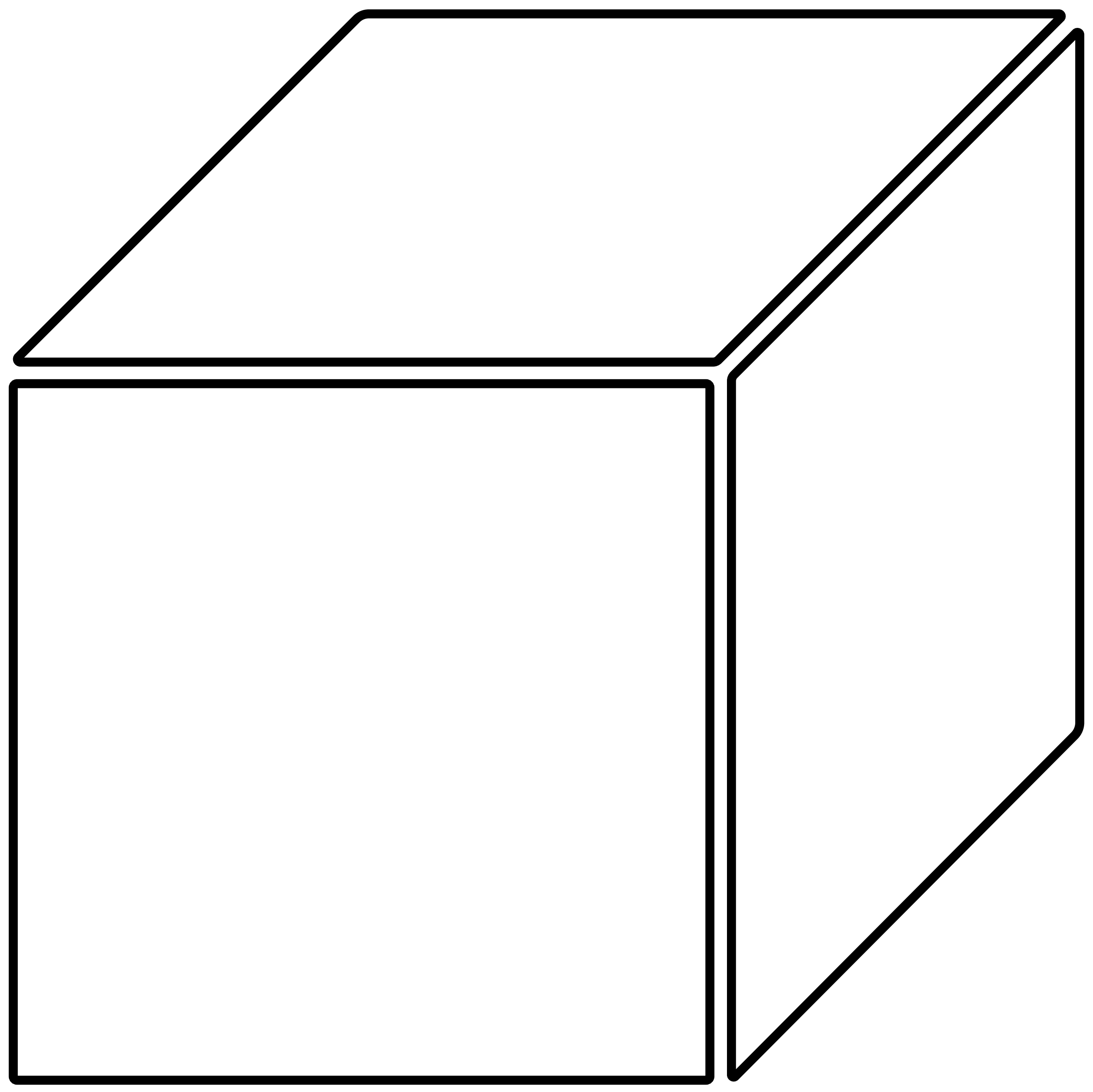
1 mm resolution

10^9 number of cells

5 channel (each 4 bytes)

20 GB storage needs for a time stamp

10^5 time steps needs 2000 TB storage



Optimum Problem

Consider a box

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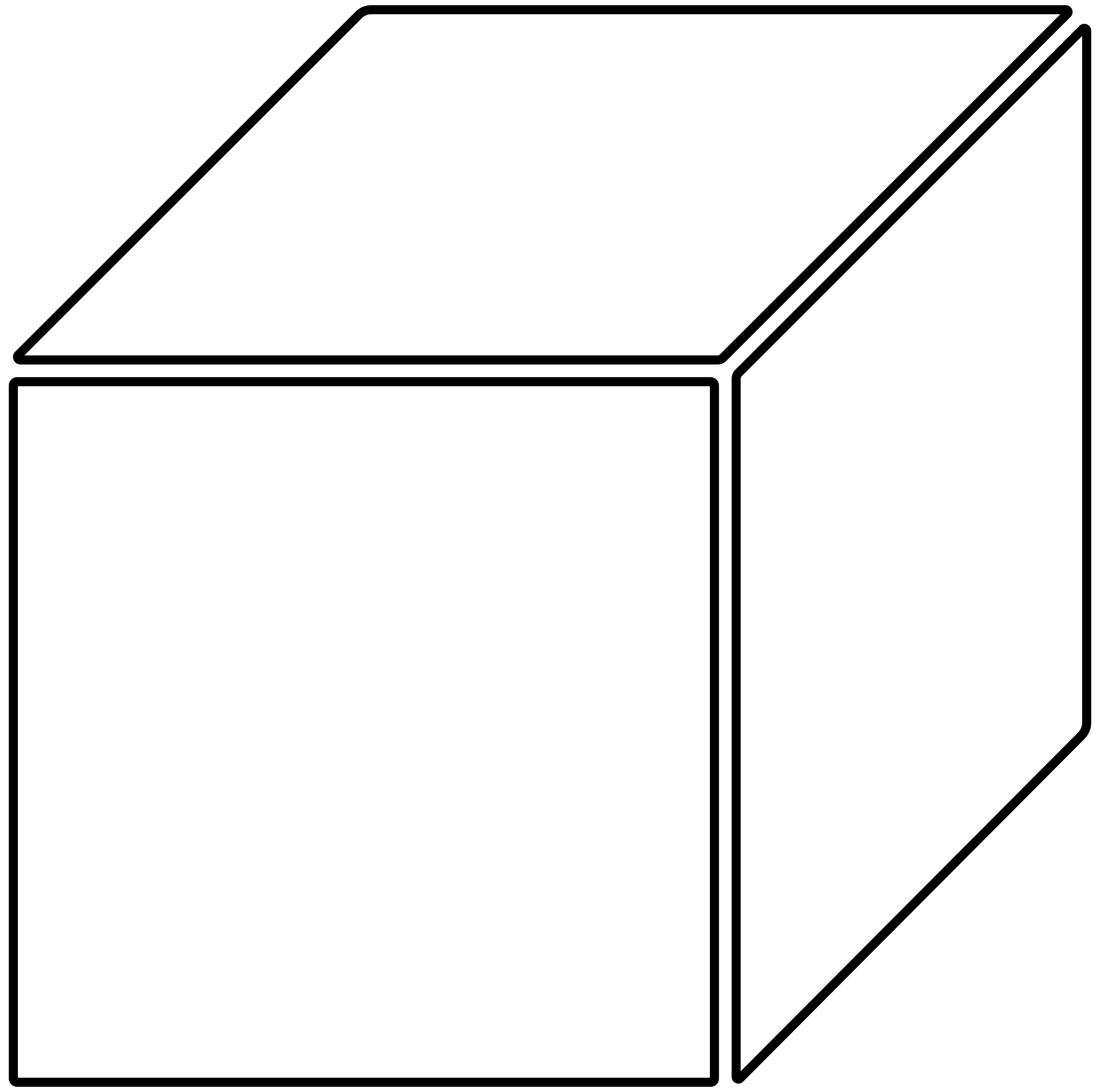

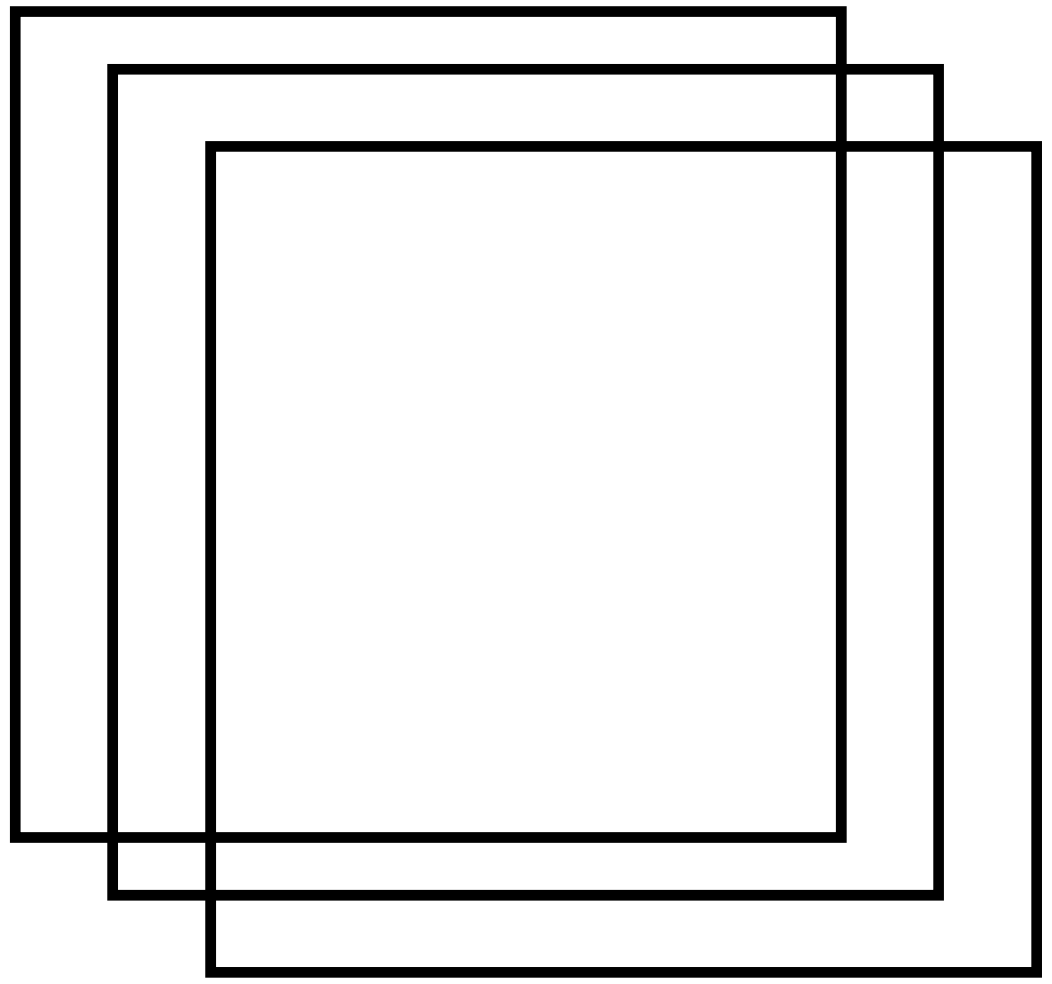

One projection images of this box is 1000×1000 pixels (20 MB)

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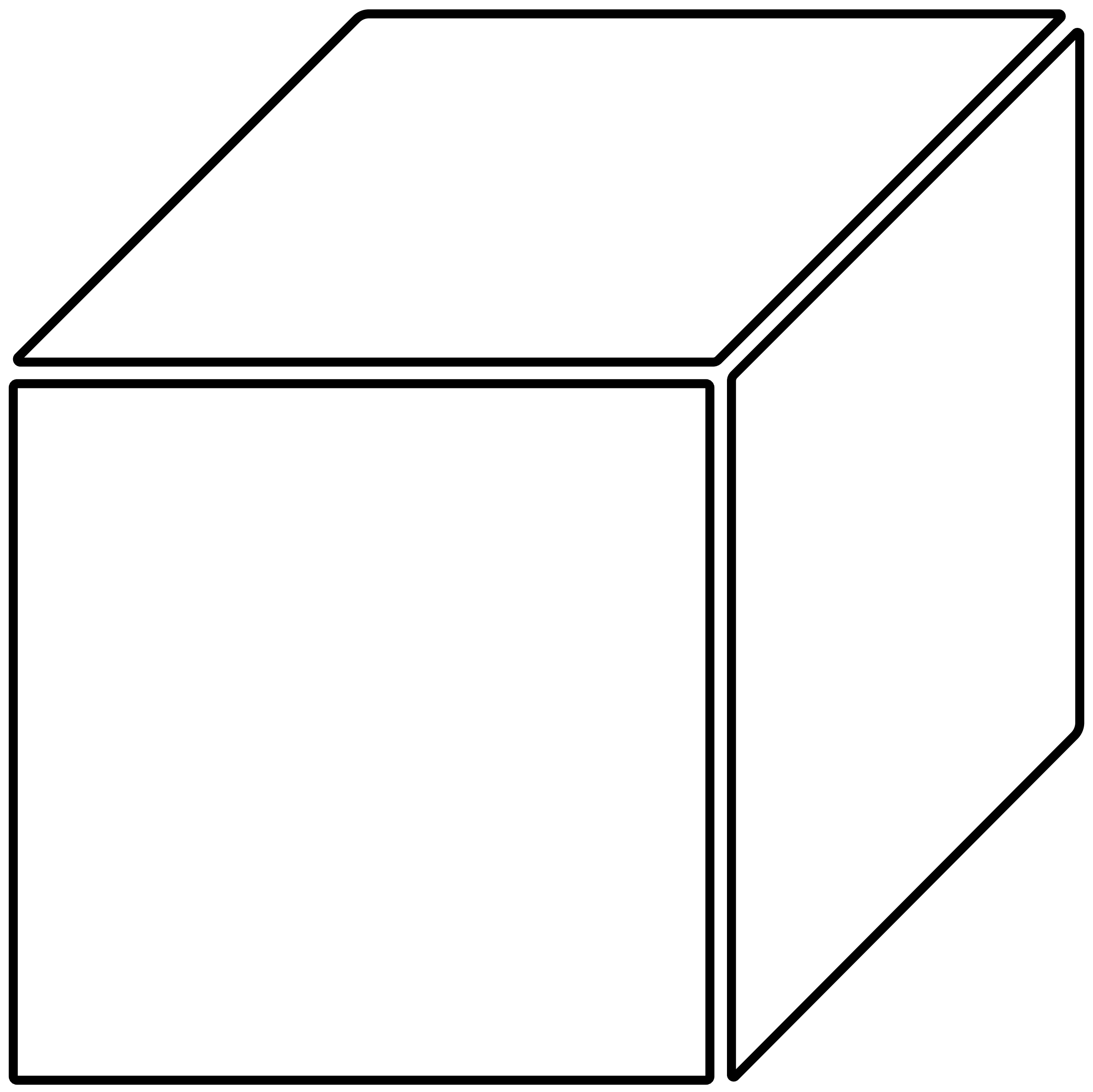
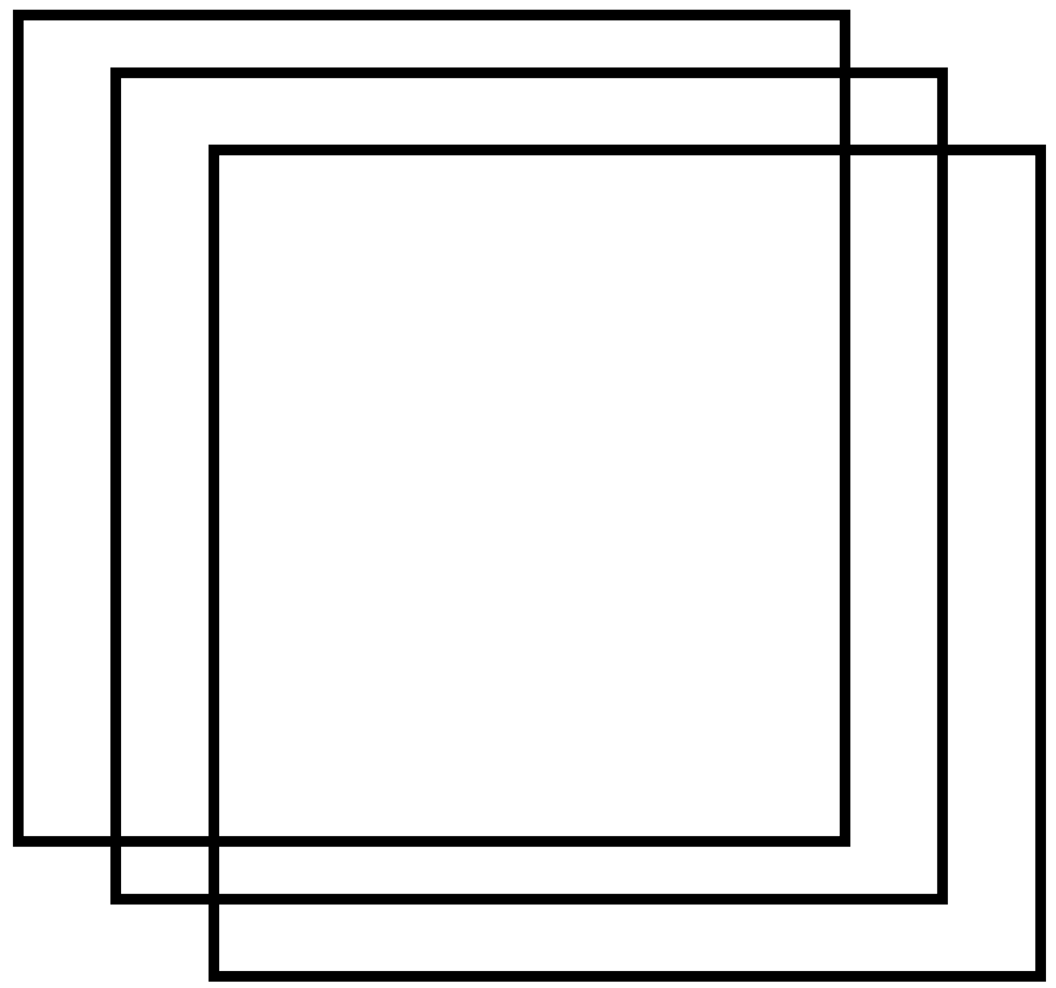
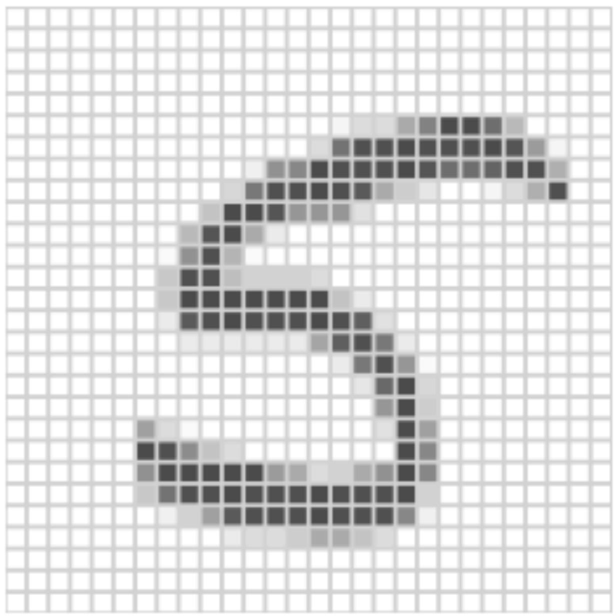




One projection images of this box is 1000 $\times$ 1000 pixels (20 MB)

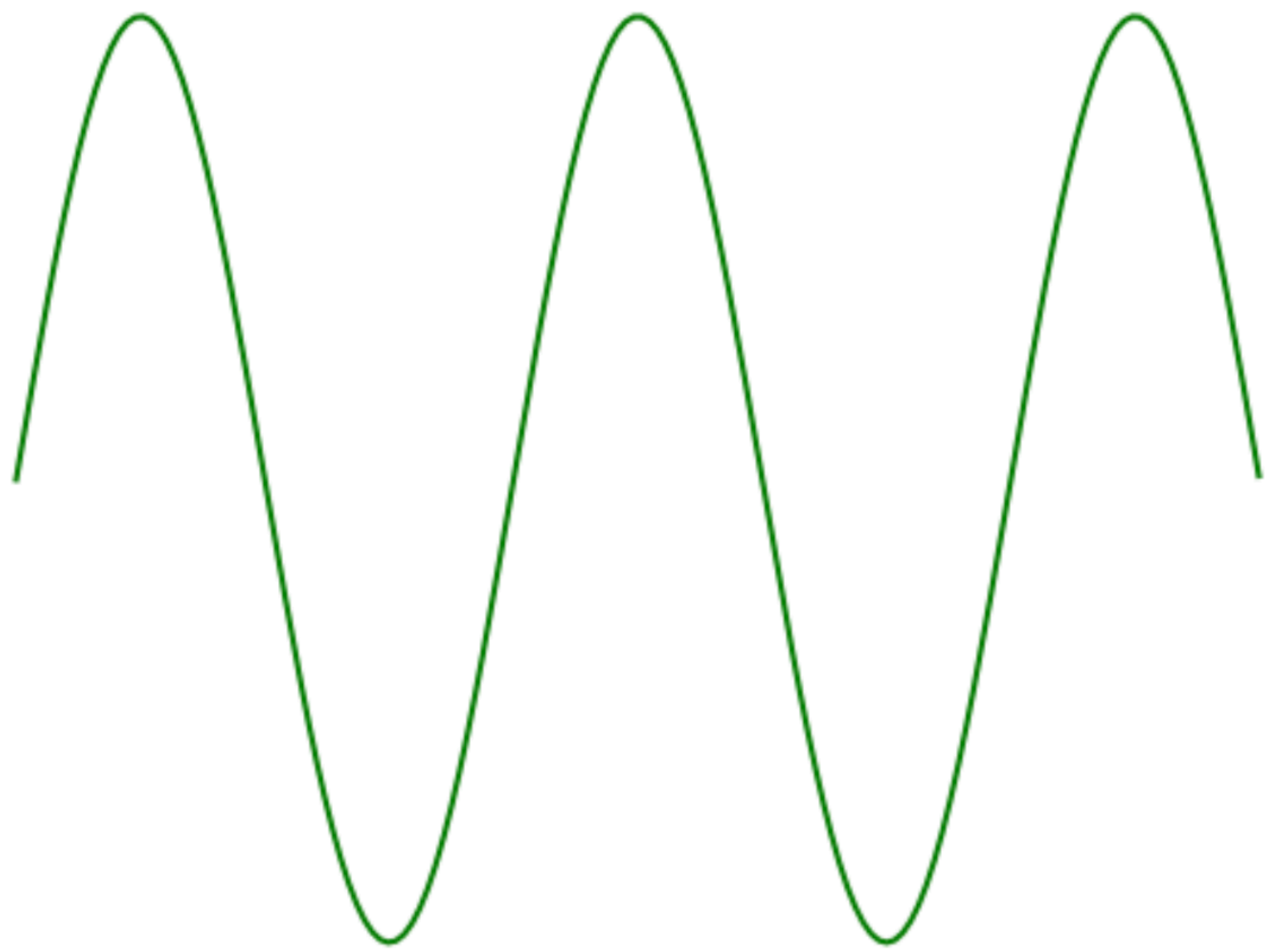
Is it possible to reconstruct the 3D of the Box
with sufficient number of 2D projection images ?

Convolution

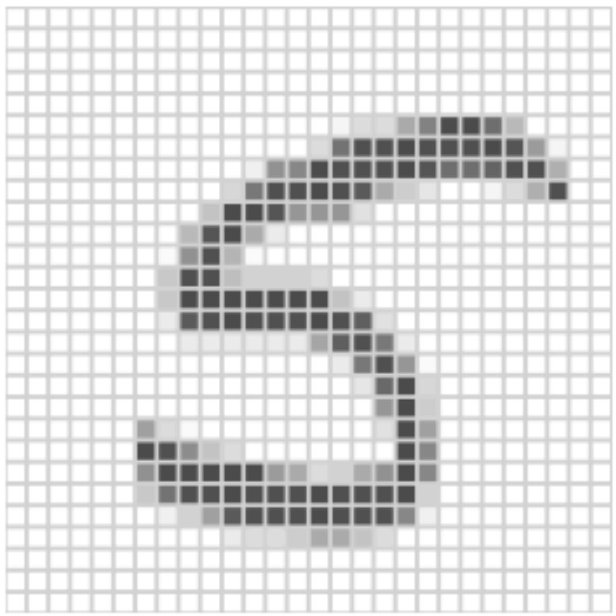


Many of the 28 $\times$ 28 pixels do not carry out information separately

Fourier analysis

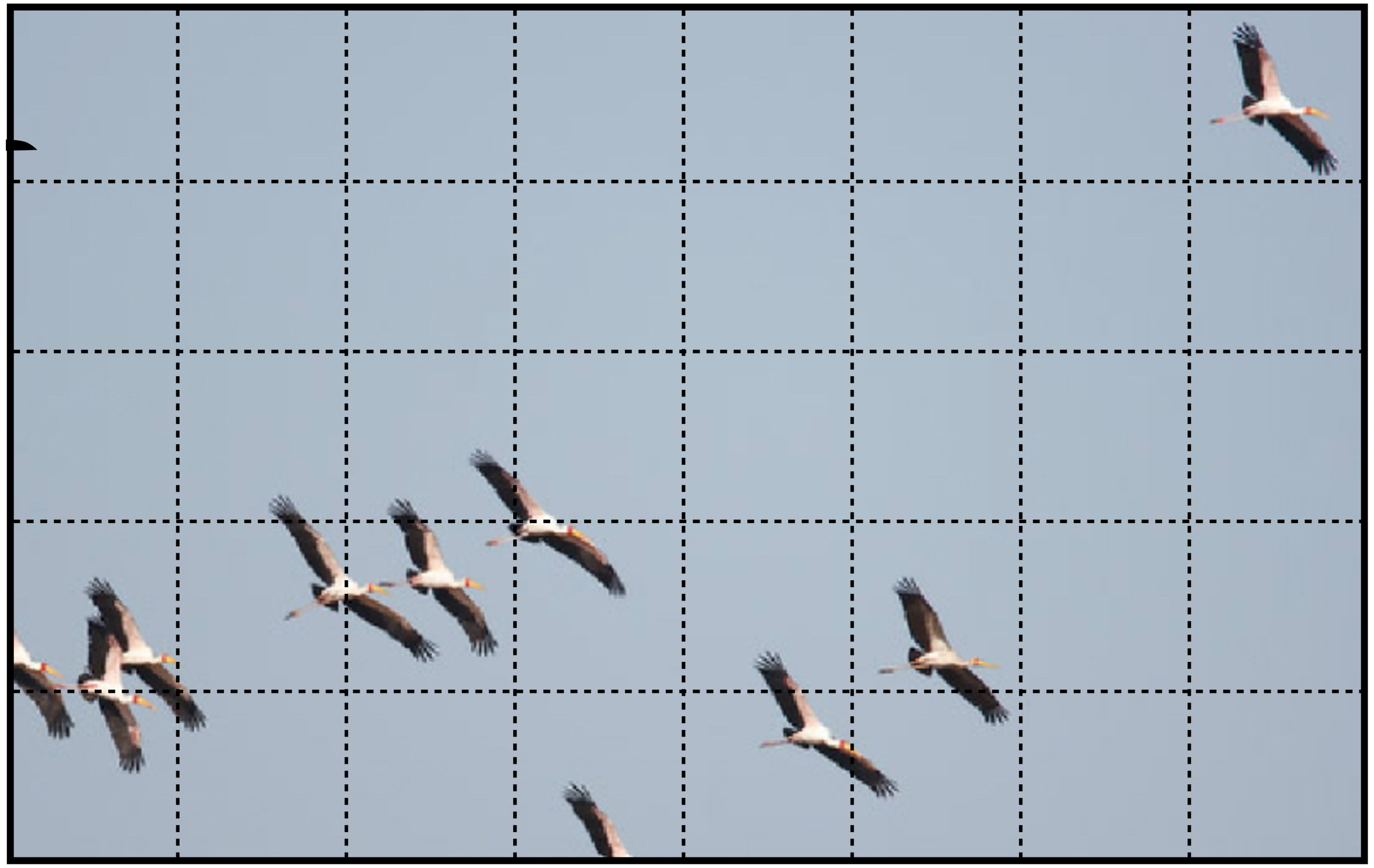


Convolution

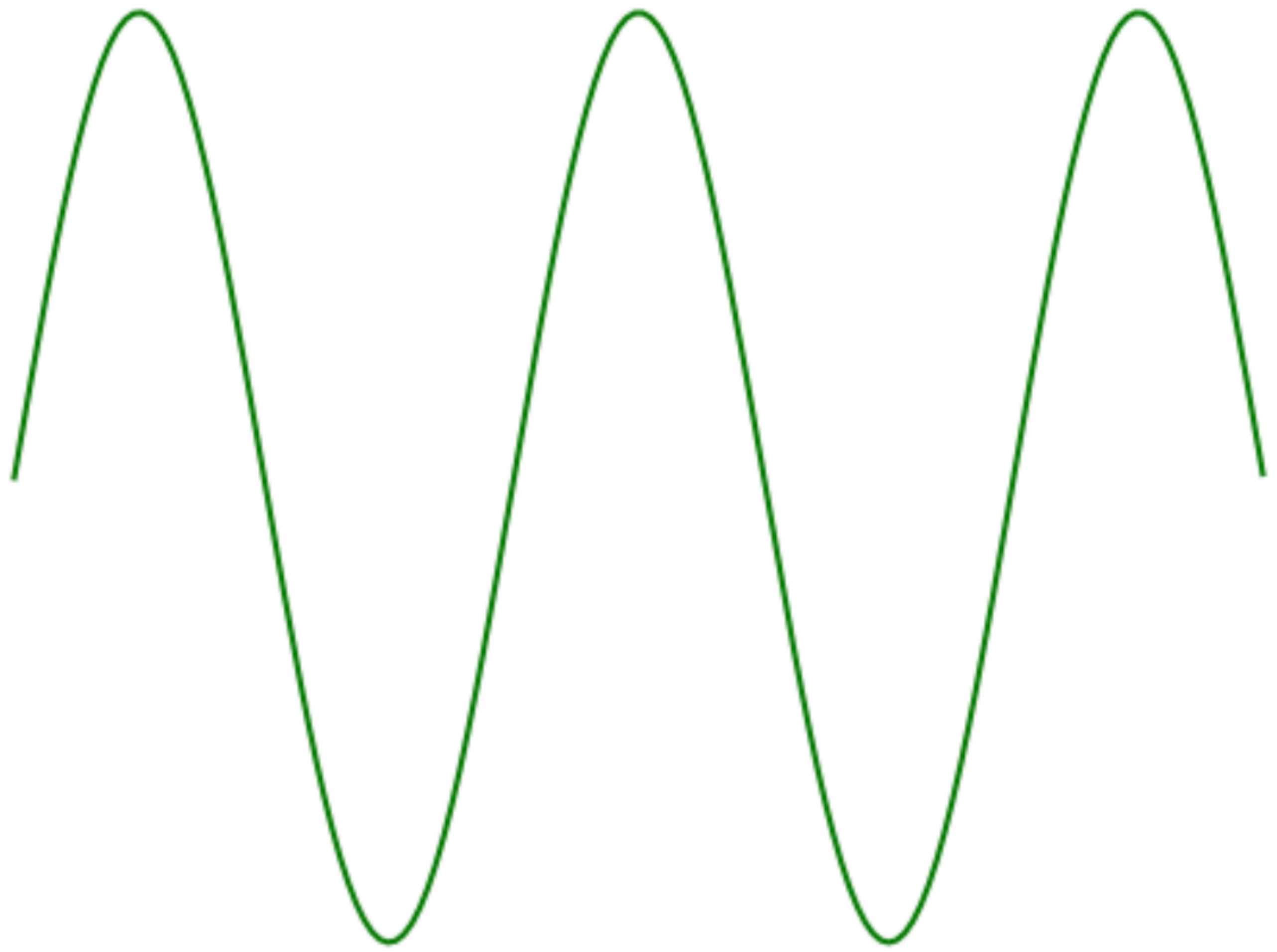


Many of the 28 $\times$ 28 pixels do not carry out information separately

image processing

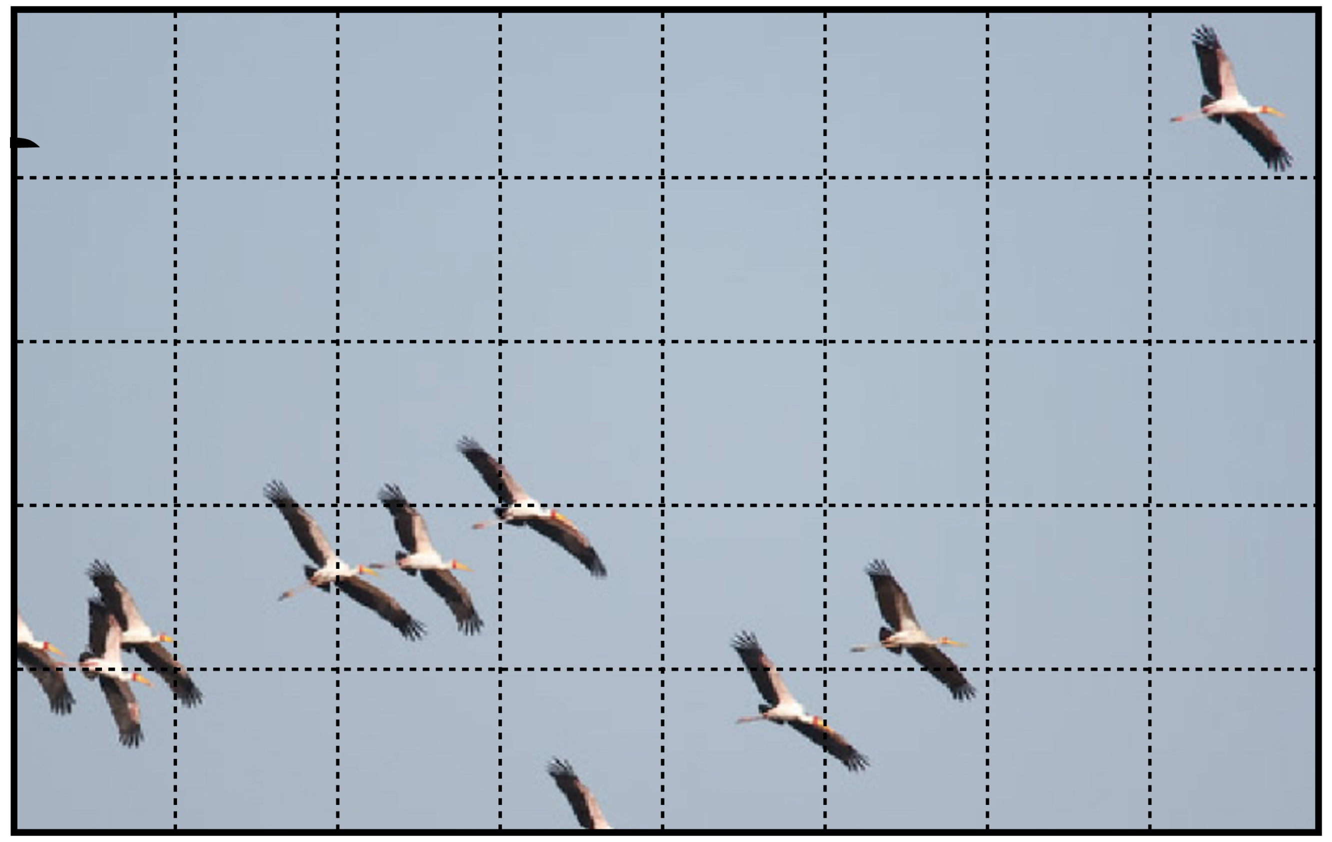


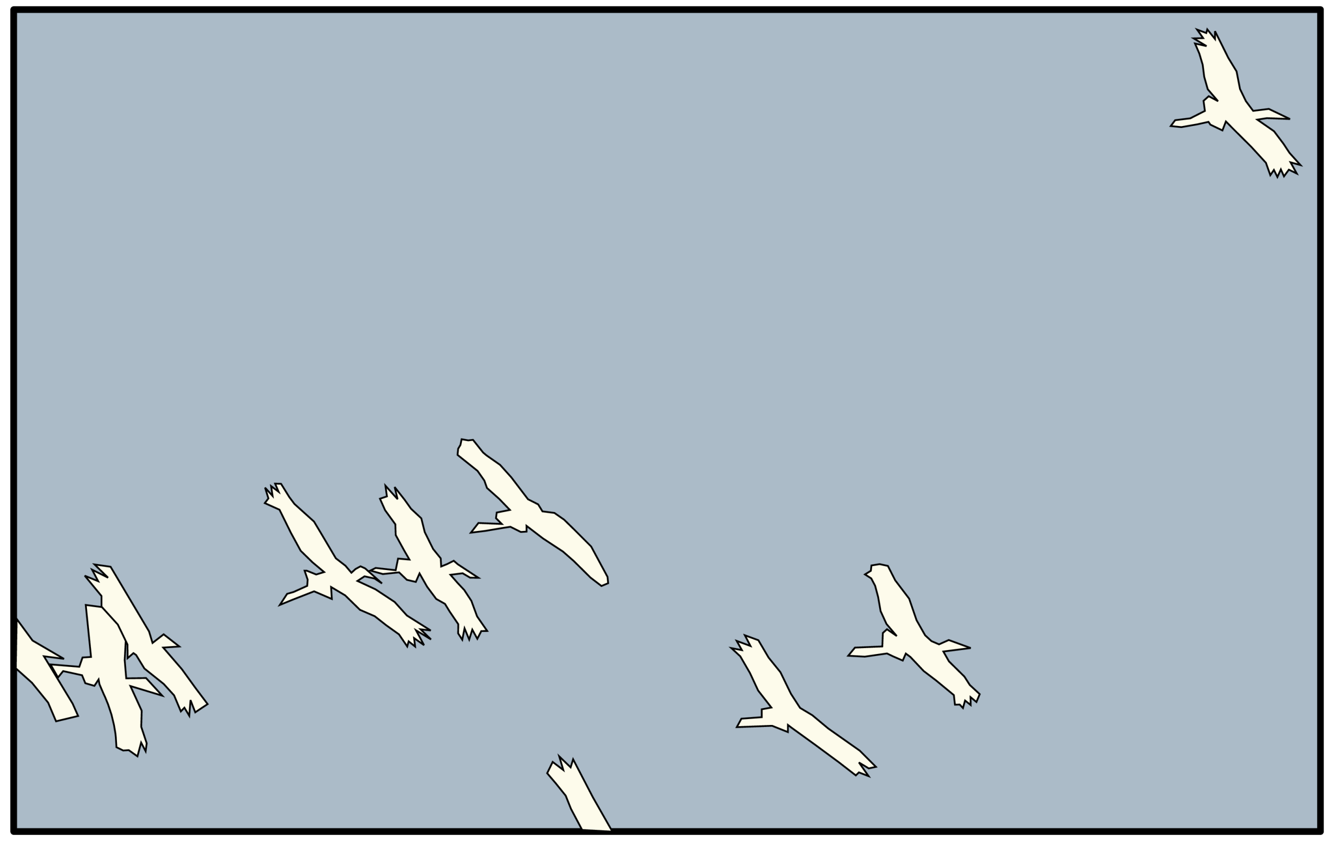
Fourier analysis



Kernel

feature map

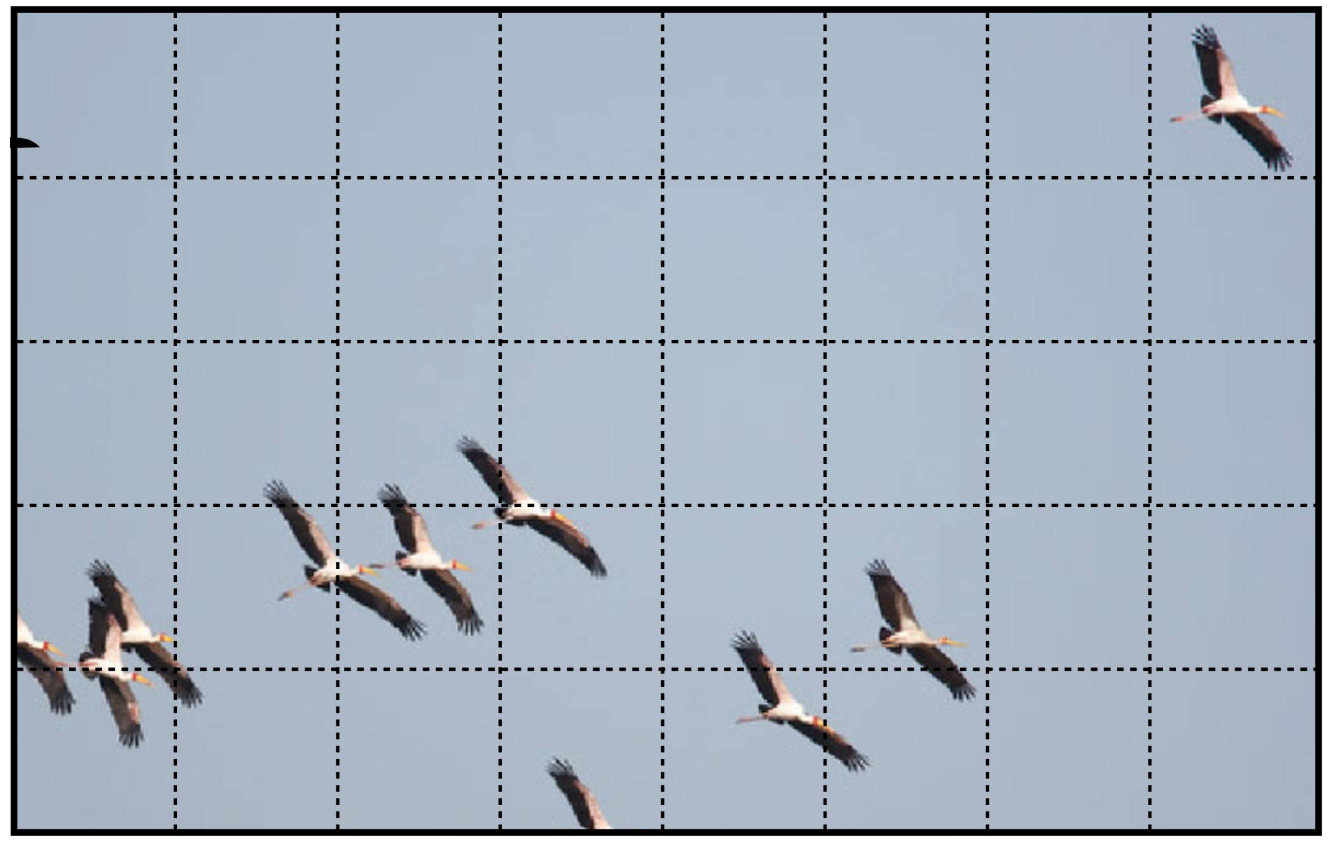


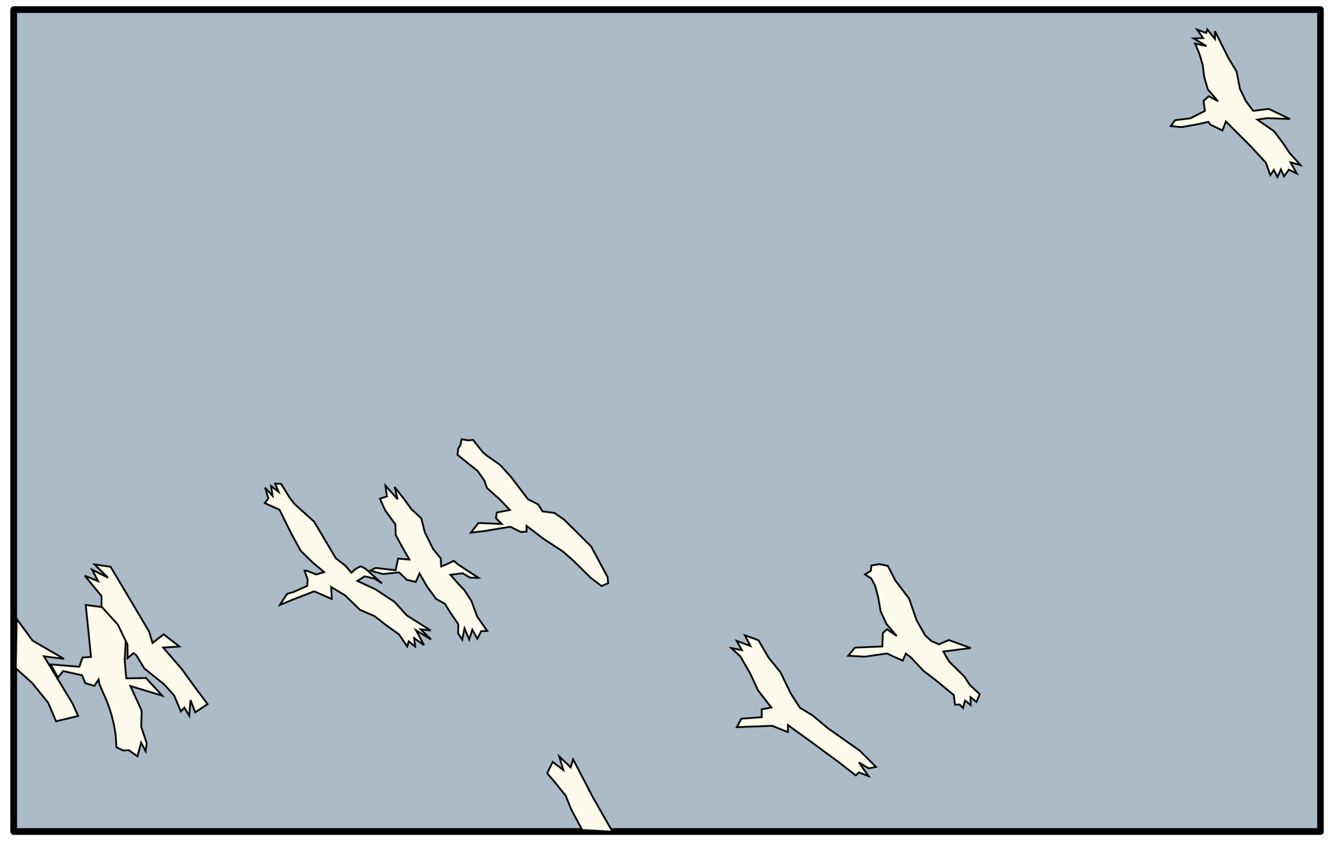


think of edge as occurring when there is a significant local change in the intensity between pixels as we move across the image.

Kernel

feature map





think of edge as occurring when there is a significant local change in the intensity between pixels as we move across the image.

Vertical Edge

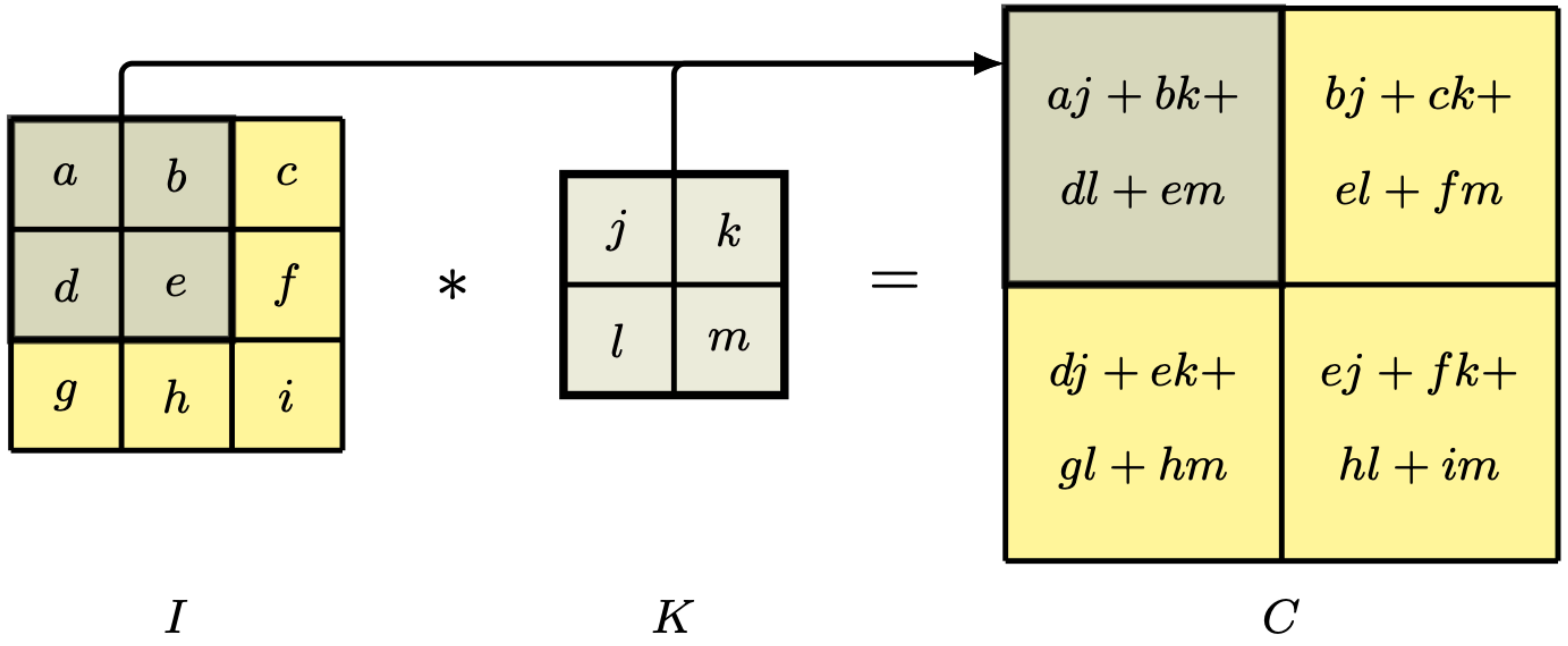
Horizontal Edge

(a) (b) (c)

-1	0	1
-1	0	1
-1	0	1

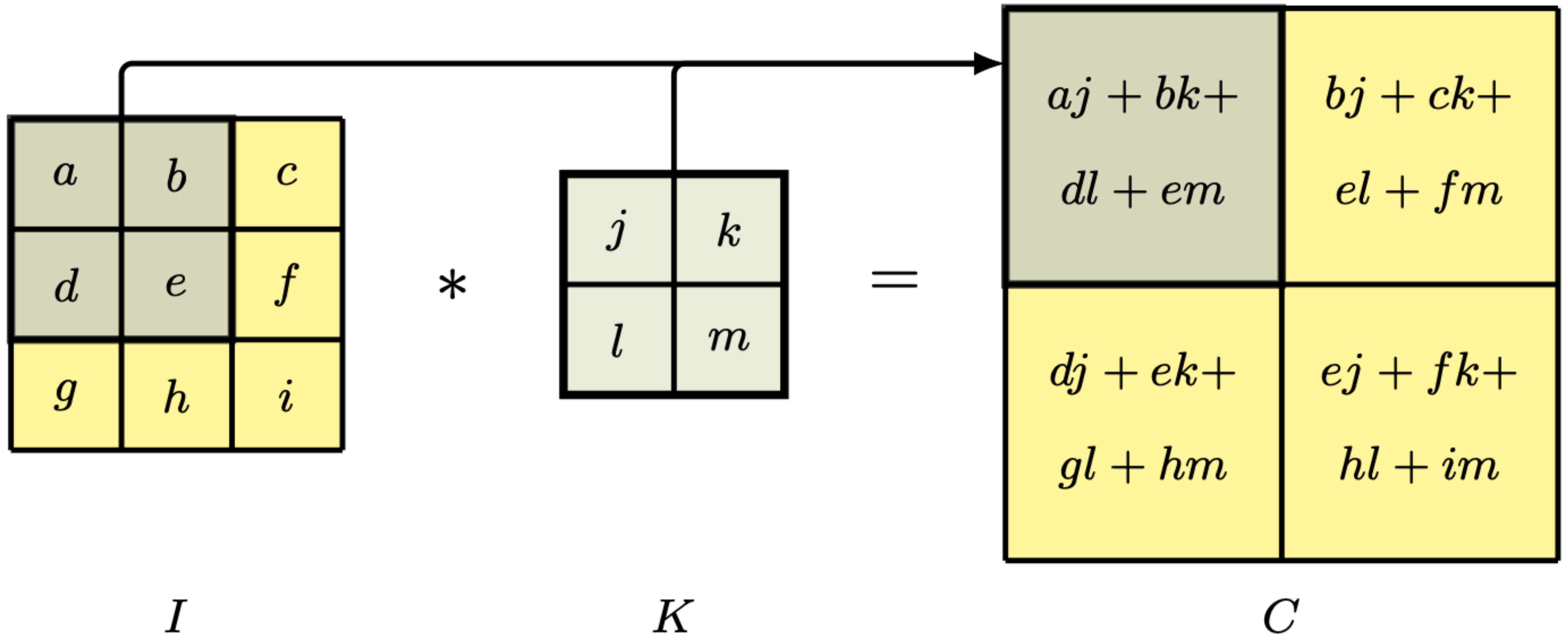
-1	-1	-1
0	0	0
1	1	1

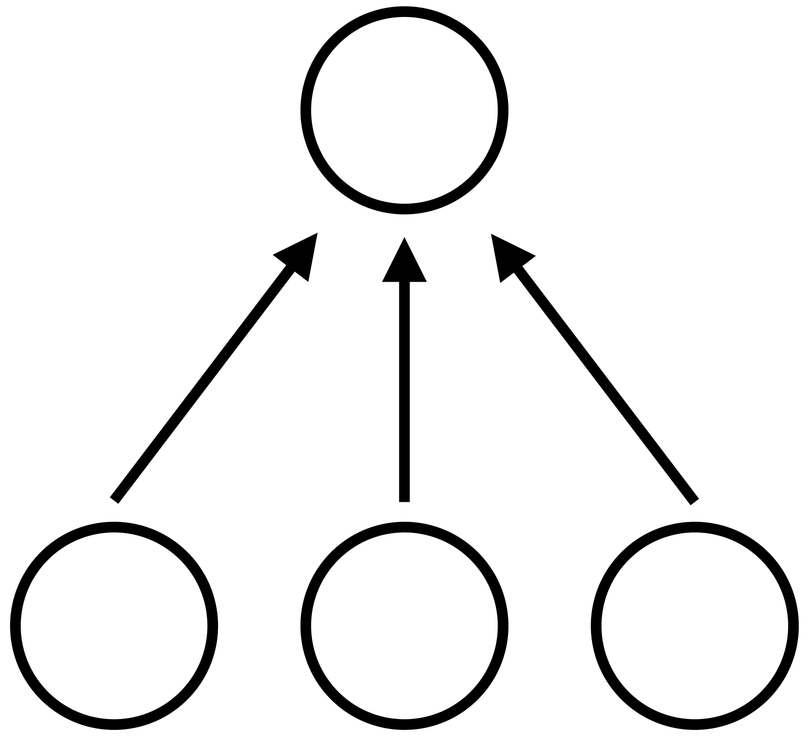
Convolution Layer

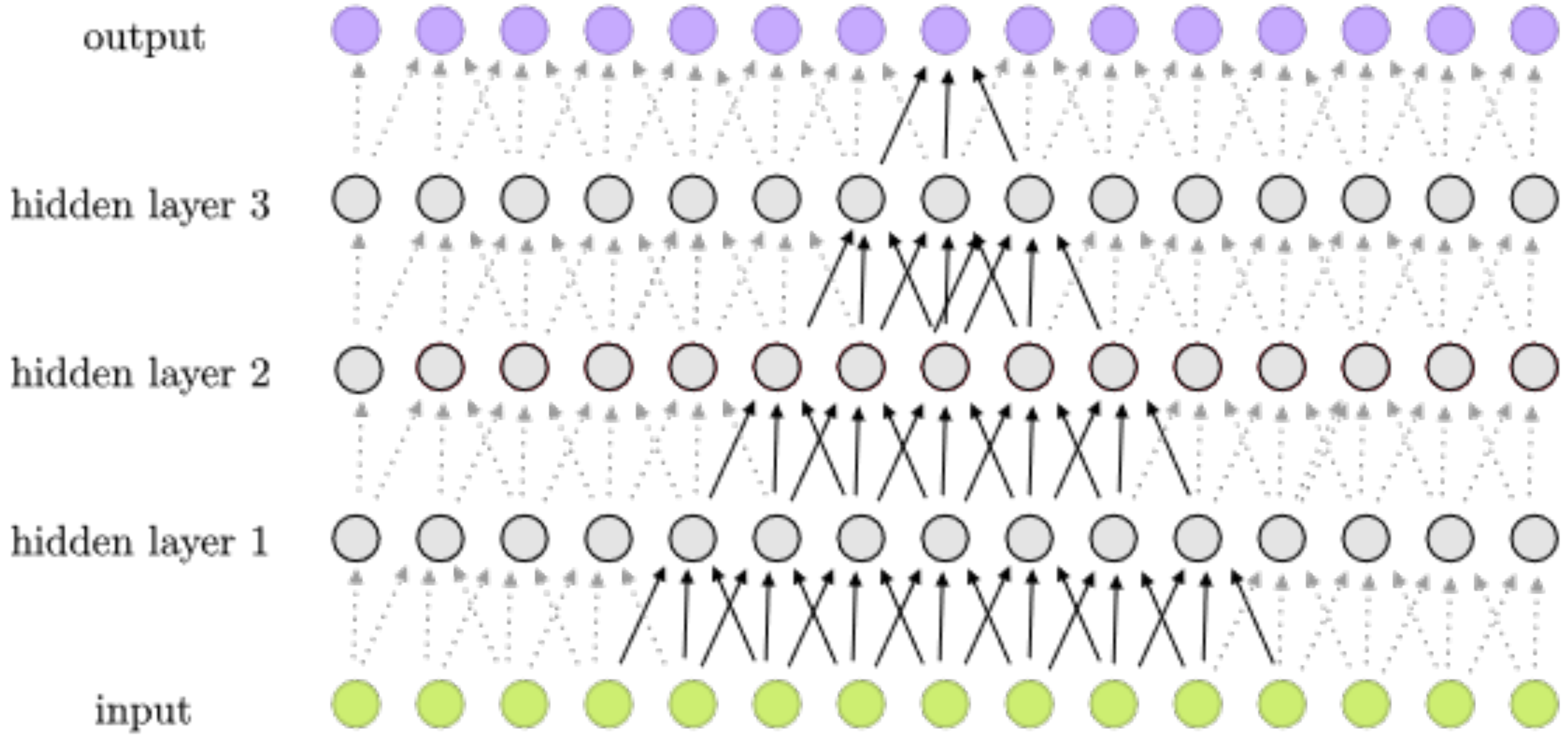


Network first identifies low-level features in the input image, such as small edges, patches of color, and the like

Convolution Layer





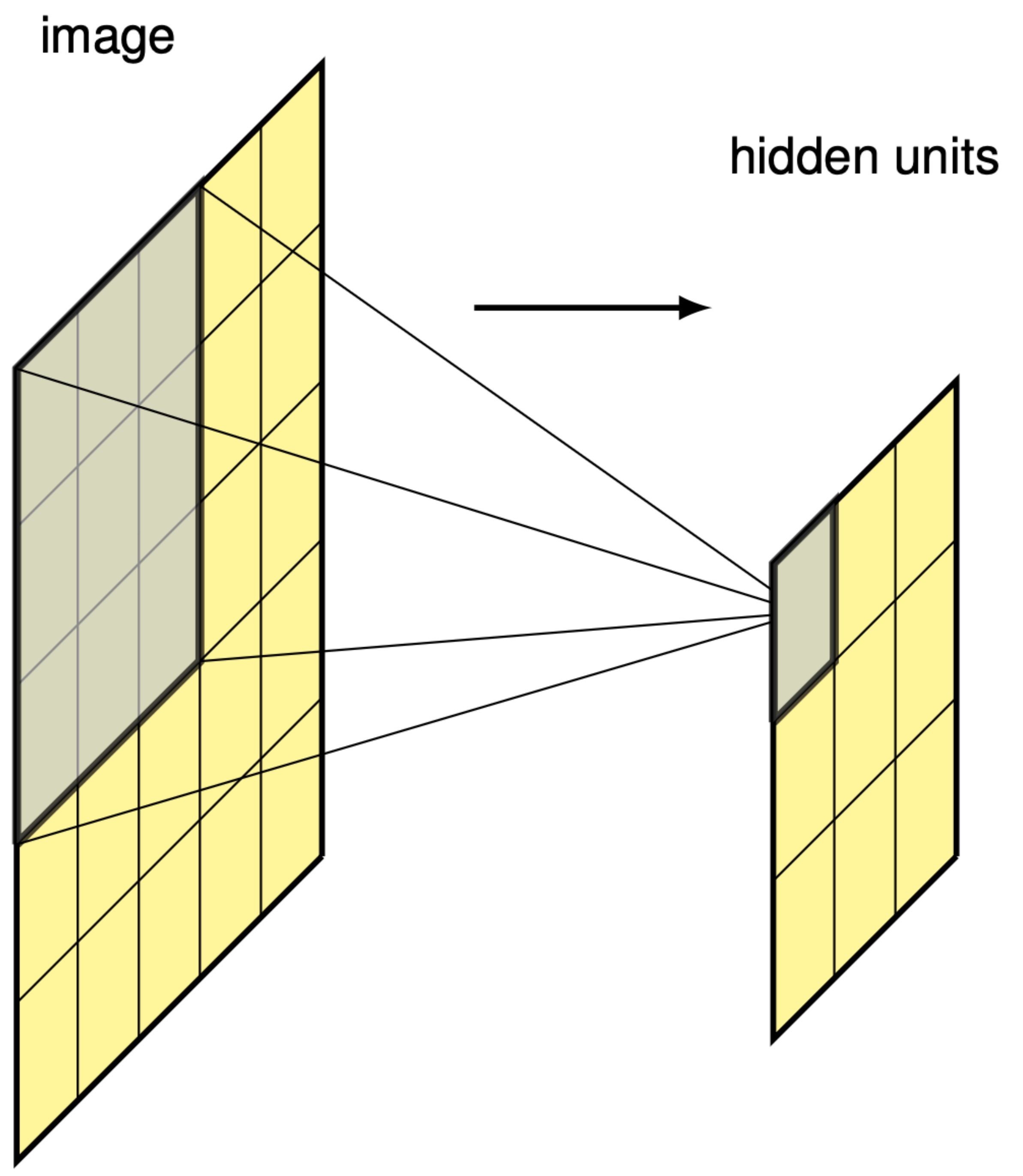


Network first identifies low-level features in the input image, such as small edges, patches of color, and the like

low-level features are combined to form higher-level features, such as parts of ears, eyes, and so on

Convolution Layer

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} * \begin{bmatrix} w_{11} & w_{12} & w_{13} \\ w_{21} & w_{22} & w_{23} \\ w_{31} & w_{32} & w_{33} \end{bmatrix} = a \, . \, w$$



(a)

maximum output response occurs when kernel detects a patch of image that, up to an overall scaling, looks like the kernel image.

Padding

typical choice is to set the padding values to mean value of the pixels

Single Channel Padded Image

0	0	0	0	0	0	0	0
0	5	0	8	7	8	1	0
0	1	9	5	0	7	7	0
0	6	0	2	4	6	6	0
0	9	7	6	6	8	4	0
0	8	3	8	5	1	3	0
0	7	2	7	0	1	0	0
0	0	0	0	0	0	0	0

1 x 1 x (6+2) x (6+2)

*

Filter

0	1	0
1	-4	1
0	1	0

1 x 1 x 3 x 3

=

Result

-19	22	-20	-12	-17	11
16	-30	-1	23	-7	-14
-14	24	7	-2	1	-7
-15	-10	-1	-1	-15	1
-13	13	-11	-5	13	-7
-18	9	-18	13	-3	4

1 x 1 x 6 x6

Region

9	5	0
0	2	4
7	6	6

x

Filter

0	1	0
1	-4	1
0	1	0

=

Element-wise Multiplication

0	5	0
0	-8	4
0	6	0

adds up to 7

Stride

$$(m, l) * (j, j) = (m - j + 1, l - j + 1)$$

$$(m - j + 1 + 2p, l - j + 1 + 2p)$$

Valid convolution

$$p = 0$$

Same convolution

$$p = (j - 1) / 2$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & X_{11} & X_{12} & X_{13} & X_{14} & 0 \\ 0 & X_{21} & X_{22} & X_{23} & X_{24} & 0 \\ 0 & X_{31} & X_{32} & X_{33} & X_{34} & 0 \\ 0 & X_{41} & X_{42} & X_{43} & X_{44} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Stride

$$(m\ ,\ l) * (j\ ,\ j) = (m - j + 1\ ,\ l - j + 1)$$

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Valid convolution

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Same convolution

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$$(\ [m - j + 2p] / s - 1\ ,\ [l - j + 2p] / s - 1\ )$$

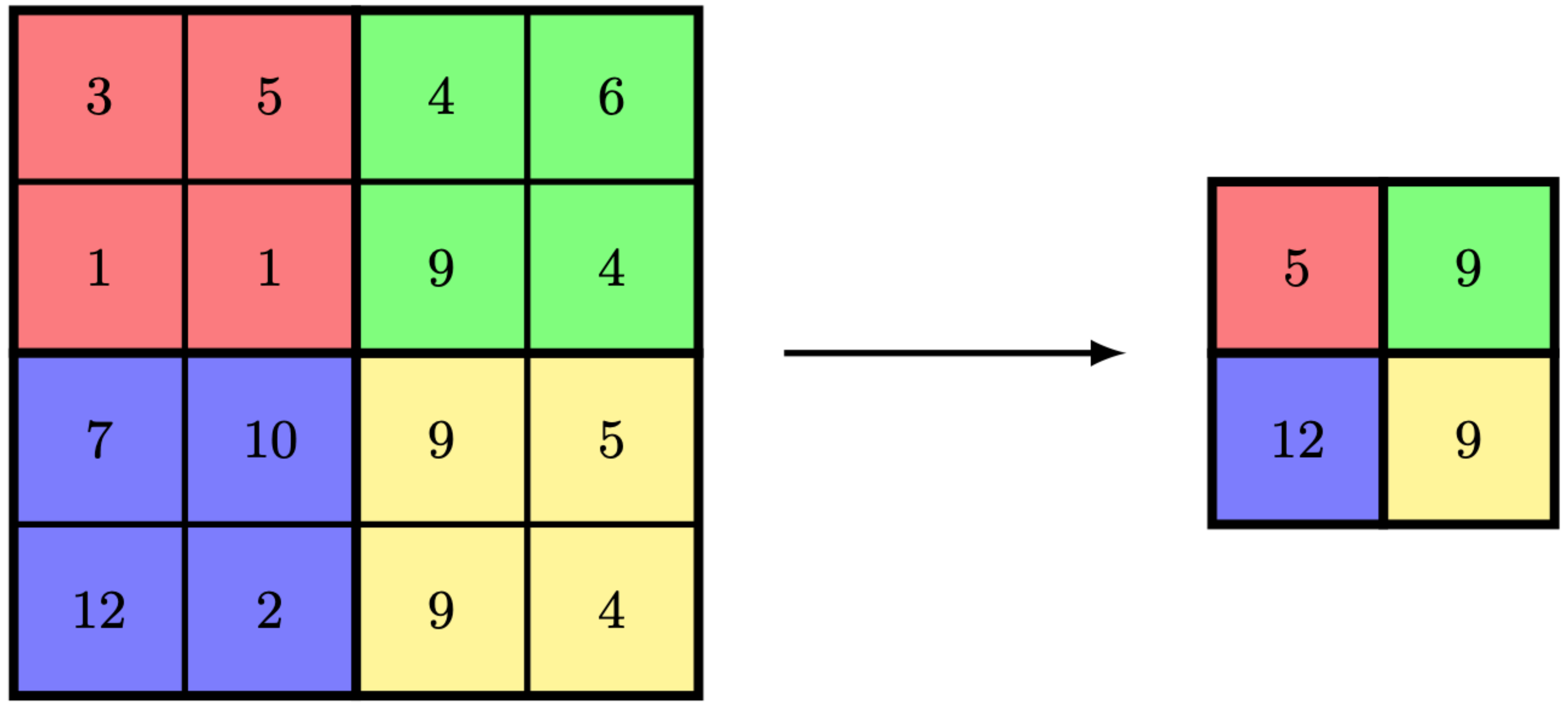
0	0	0	0	0	0
0	X_{11}	X_{12}	X_{13}	X_{14}	0
0	X_{21}	X_{22}	X_{23}	X_{24}	0
0	X_{31}	X_{32}	X_{33}	X_{34}	0
0	X_{41}	X_{42}	X_{43}	X_{44}	0
0	0	0	0	0	0

Convolutional Neural Network

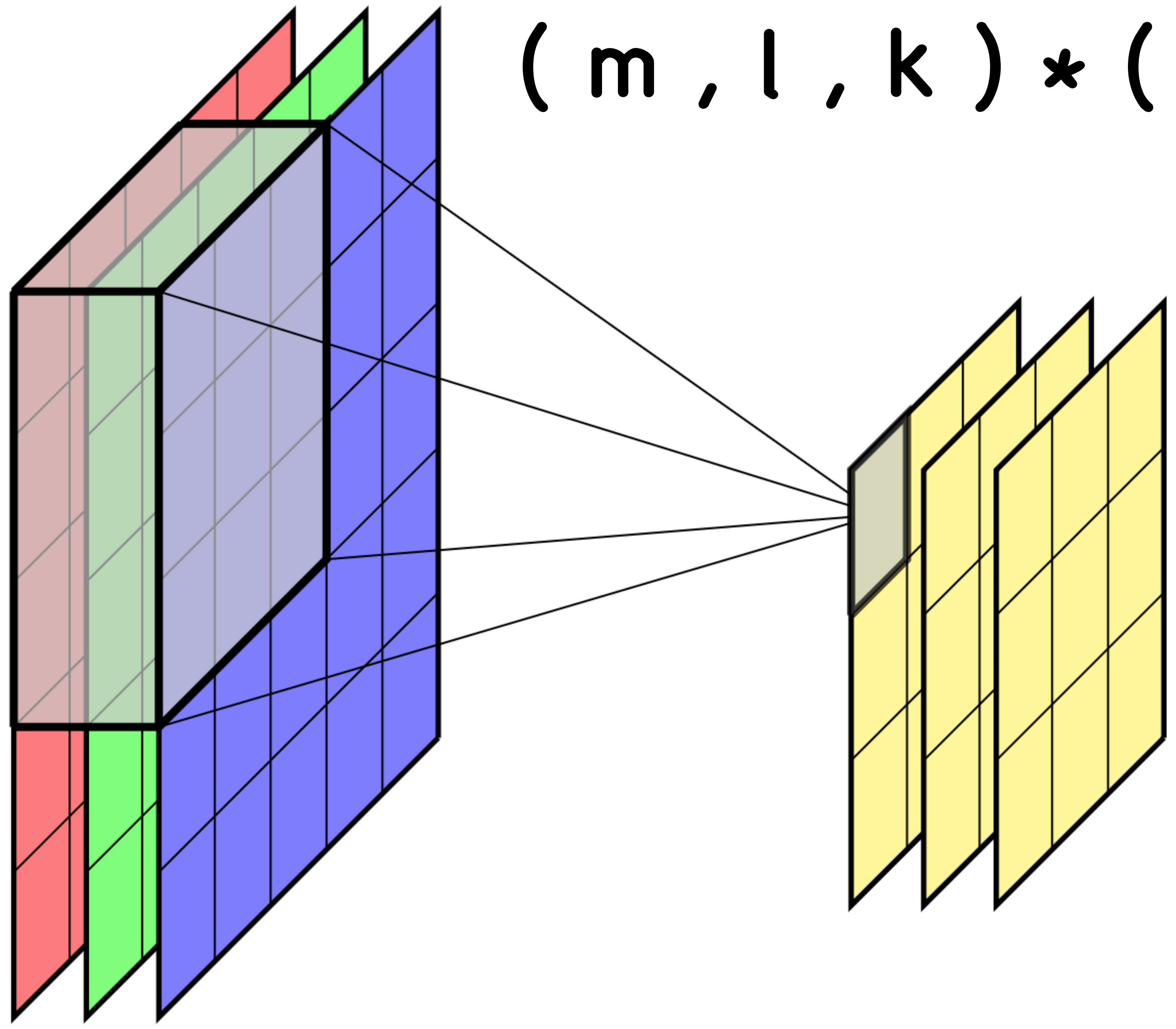
combines two types of hidden layers : 1)convolution layers 2) pooling layers

Convolution layers search for instances of small patterns in the image, whereas pooling layers downsample these to select a prominent subset.

Max Pooling - Average Pooling



$$(m , l , k) * (j , j , k , k_{out})$$



Count number of weight ?

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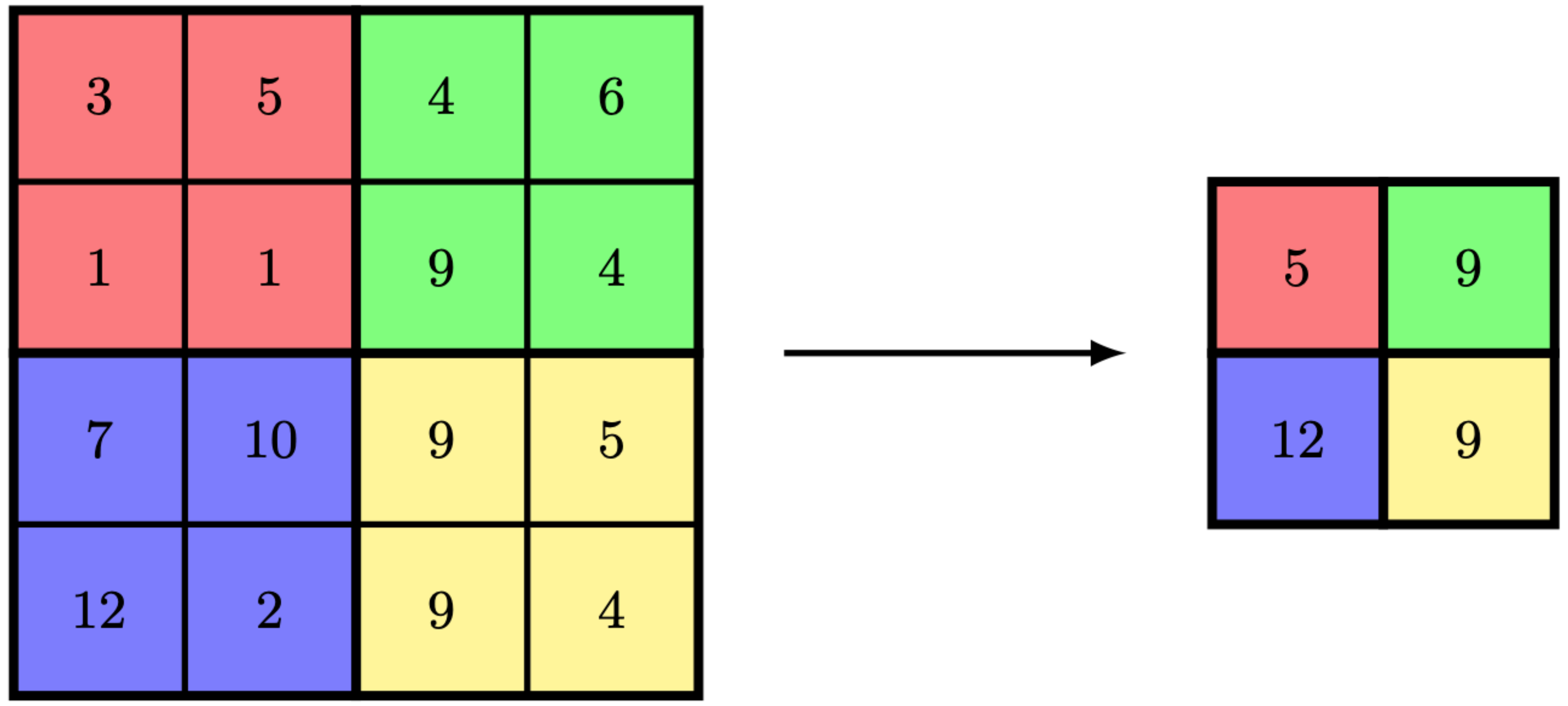
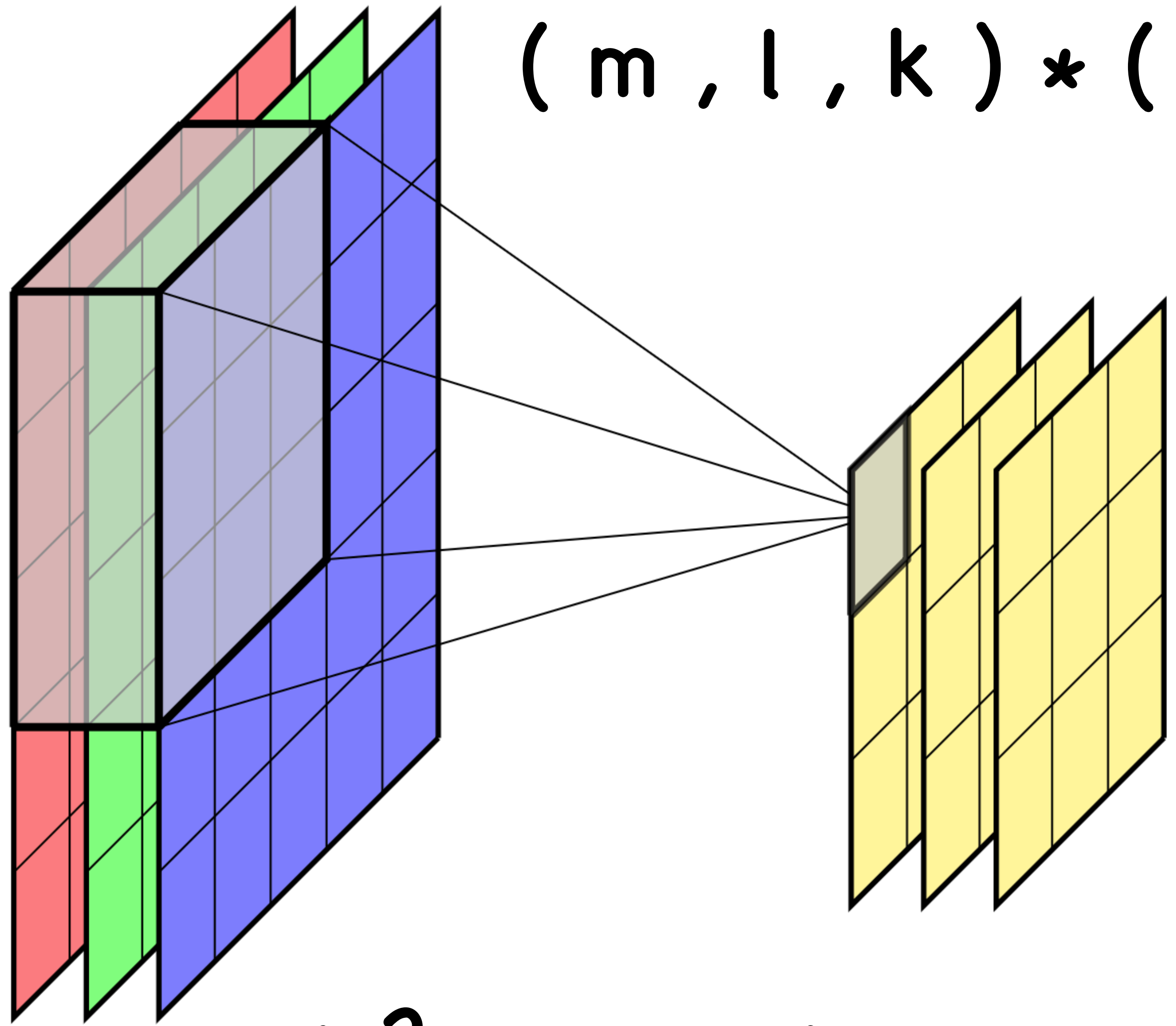


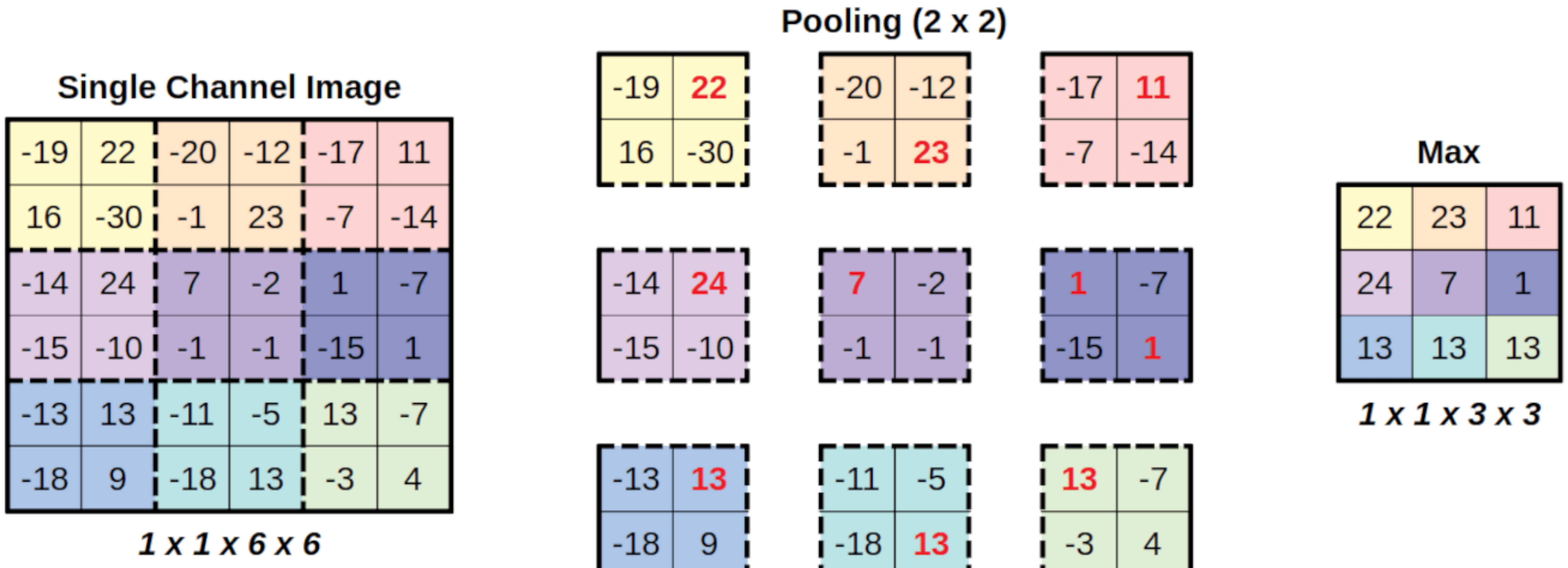
$(m, l, k) * (j, j, k, k_{out})$



$(j^2 \times k + 1)\, k_{out}$

Number of k_{out} filter with shape (j, j, k)

Pooling



CNN Architecture

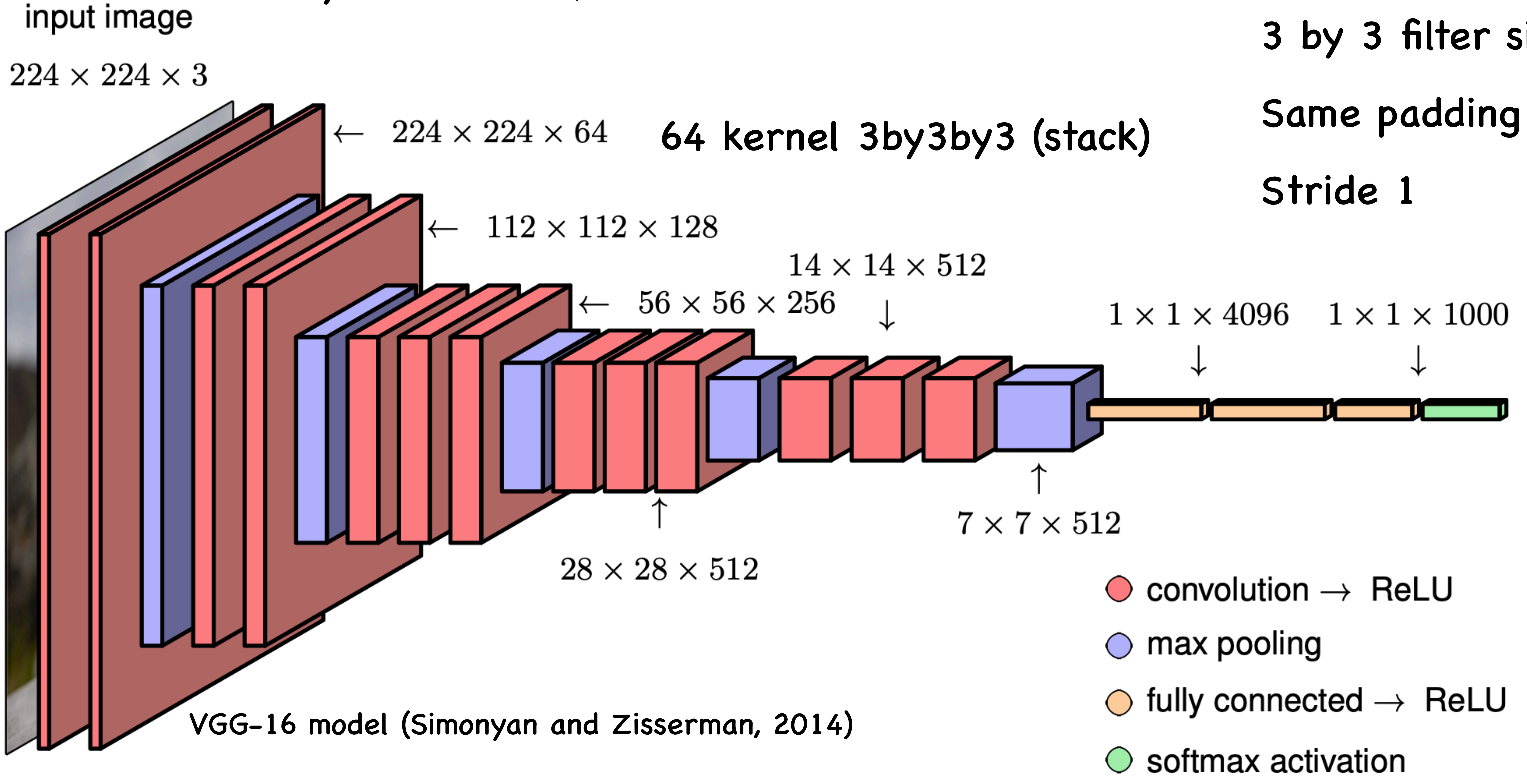
2 by 2 filter size, Stride 2

64 kernel 3by3by3 (stack)

3 by 3 filter size,

Same padding

Stride 1



VGG-16 model (Simonyan and Zisserman, 2014)

To classify 1000 categories

138 million parameters, 103 million are in the first fully connected layer

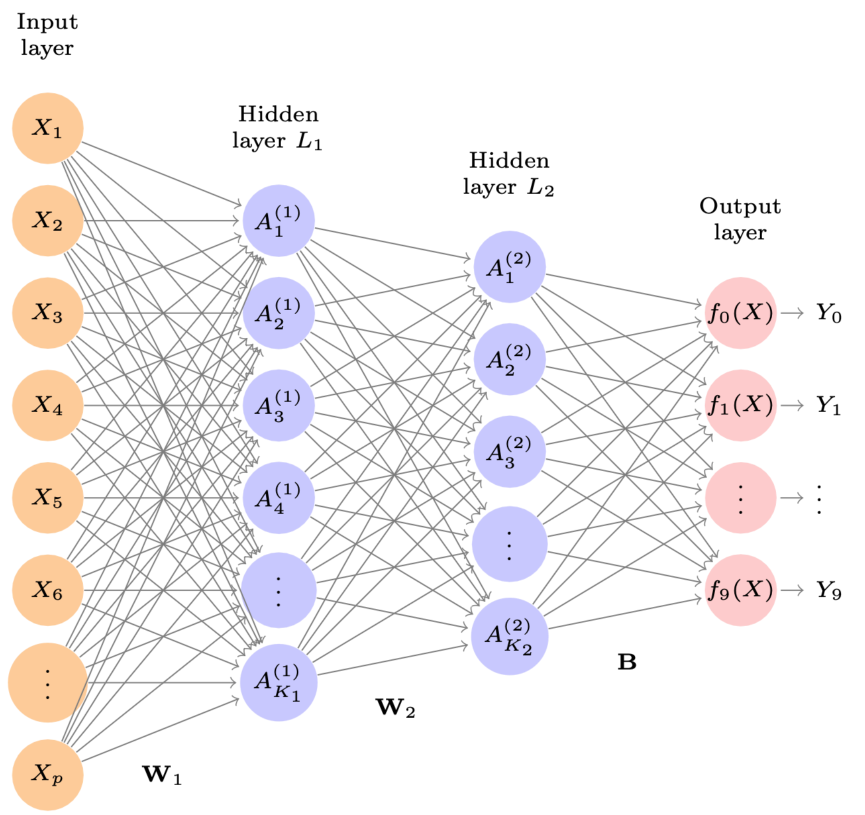
Compare with Dense network

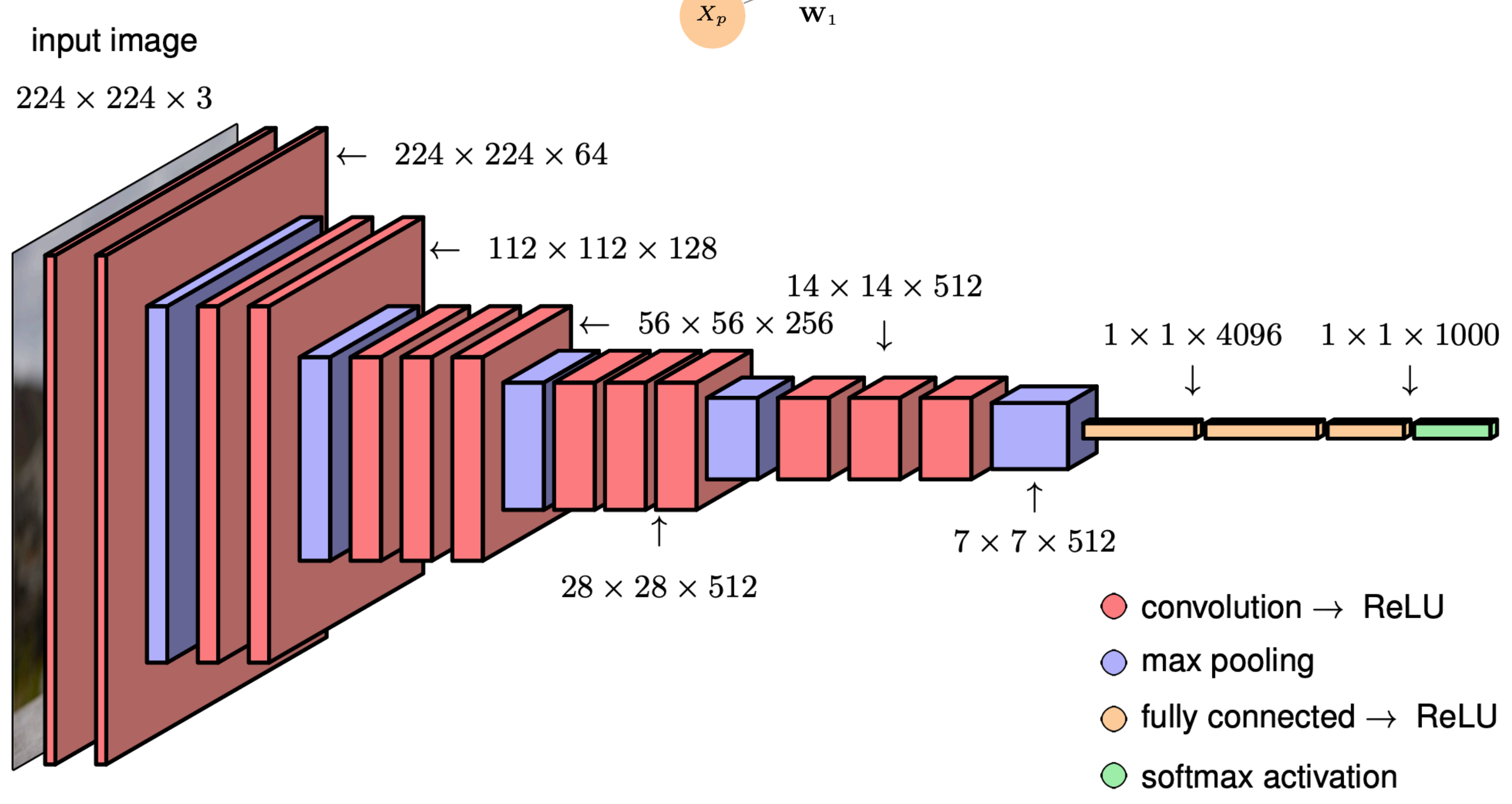
Layers encodes translation equivariance

connections are sparse, leading to far fewer weights even with large images

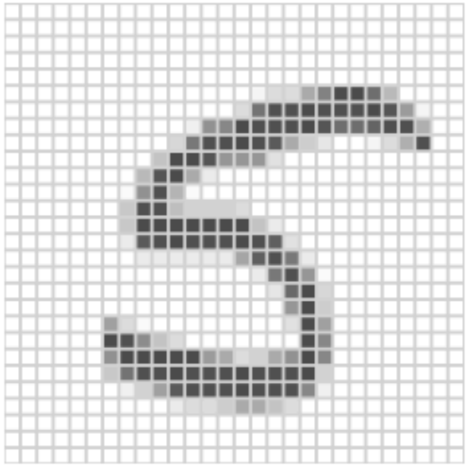
weight values are shared, greatly reducing the number of independent parameters and consequently reducing the required size of the training set needed to learn those parameters

the same network can be applied to images of different sizes without the need for retraining.





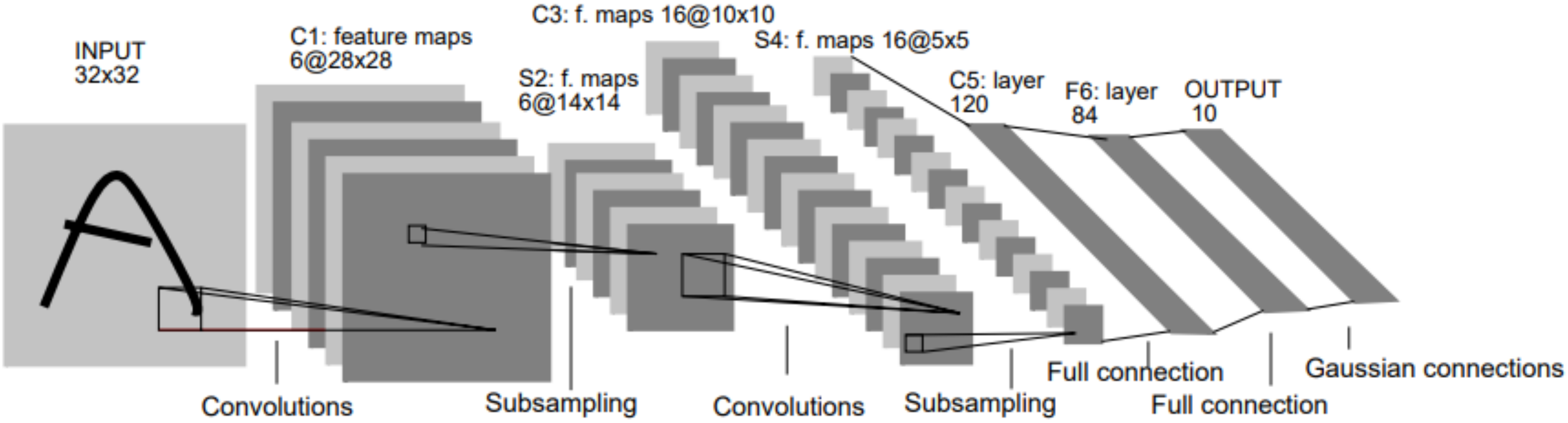
LeNet-5



LeCun, Yann, et al. "Gradient-based learning applied to document recognition." Proceedings of the IEEE 86.11 (2002): 2278-2324

More than 80,000 citation

Not only for digit also for alphabet



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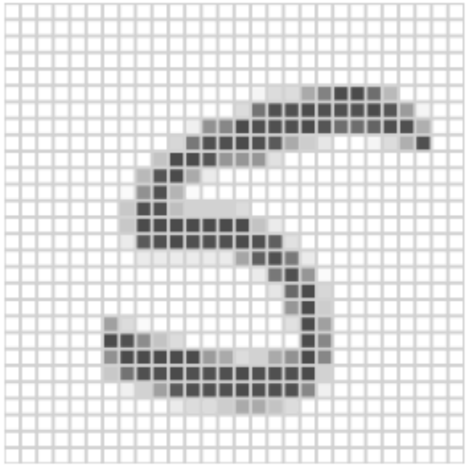
LeNet-5



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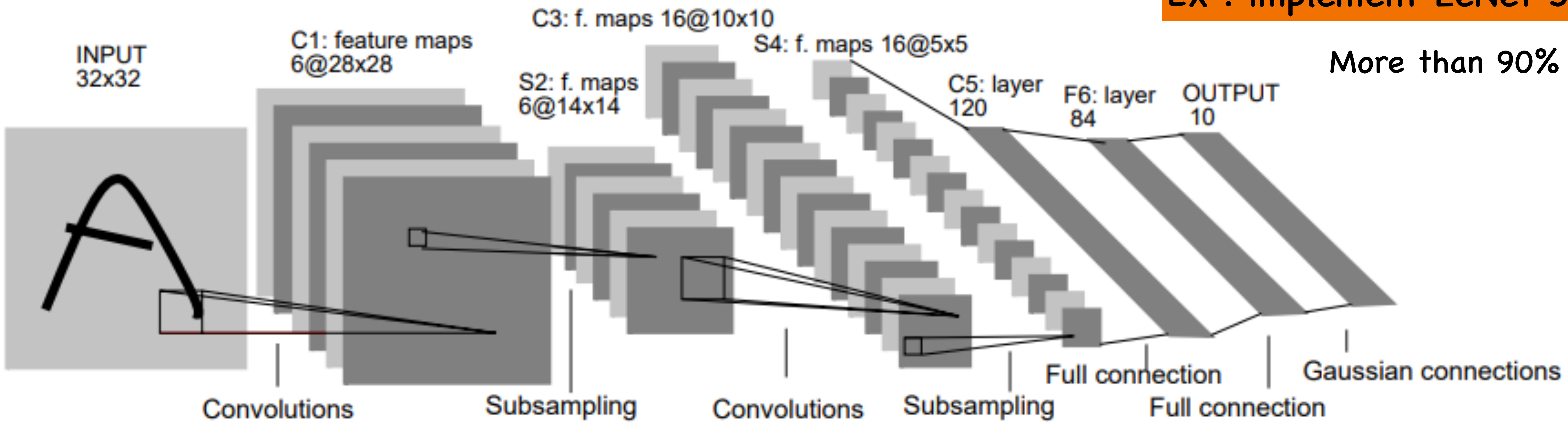
Ex : implement LeNet-5

More than 90% accuracy



ImageNet

AI researcher Fei-Fei Li began working on the idea for ImageNet in 2006.

At a time when most AI research focused on models and algorithms, Li wanted to expand and improve the data available to train AI algorithms.

In 2007, Li met with Princeton professor Christiane Fellbaum, to discuss the project.

49K workers from 167 countries filtering and labeling over 160M candidate images

14 million images labelled three times

final dataset spans 1000 object classes and contains 1,281,167 training images, 50,000 validation images and 100,000 test images.

https://www.image-net.org/download.php

ImageNet Lesson

MNIST Lesson

How many images can by classified by a normal person ?

ImageNet Lesson

MNIST Lesson

How many images can by classified by a normal person ? 40,000

10,000 sample for each class

40,000 class

Total images of 400,000,000

How images were labelled ?

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ImageNet Lesson

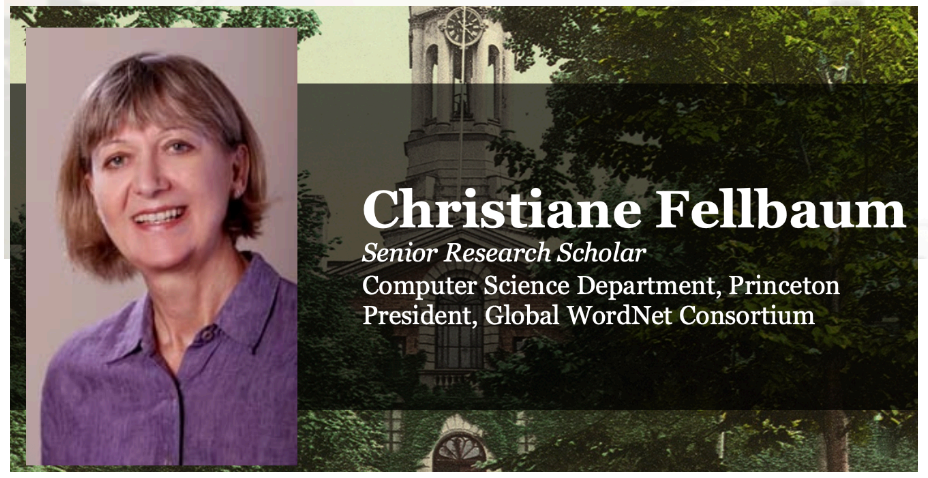
MNIST Lesson

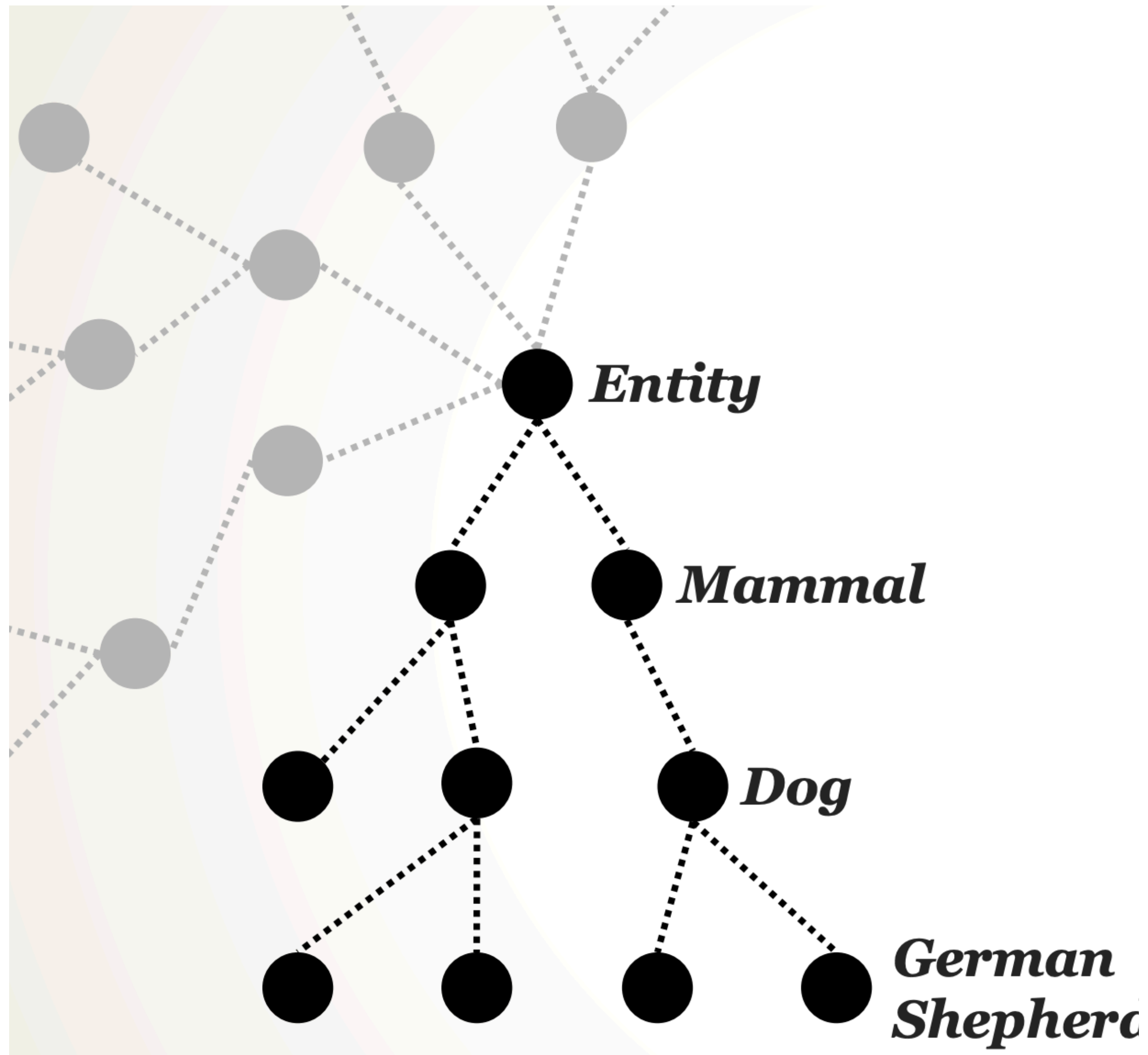
What is WordNet?

Original paper by **[George Miller, et al 1990]** cited over 5,000 times

Organizes over 150,000 words into 117,000 categories called *synsets*.

Establishes ontological and lexical relationships in NLP and related tasks.





https://image-net.org/static_files/files/imagenet_ilsvrc2017_v1.0.pdf